

New results in Share Conversion, with applications to evolving access structures

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Abstract. We say there is a share conversion from a secret-sharing scheme Π to another scheme Π' implementing the same access structure if each party can locally apply a deterministic function to their share to transform any valid secret-sharing under Π to a valid (but not necessarily random) secret-sharing under Π' of the same secret. If such a conversion exists, we say that $\Pi \geq \Pi'$. This notion was introduced by Cramer et al. (TCC'05), where they particularly proved that for any access structure, any linear secret-sharing scheme over a given field $\mathbb{F}$, has a conversion from a CNF scheme, and is convertible to a DNF scheme. In this work, we initiate a systematic study of convertability between secret-sharing schemes, and present a number of results with implications to the understanding of the convertability landscape.

- In the context of linear schemes, we present two key theorems providing necessary conditions for convertability, proved using linear-algebraic tools. It has several implications, such as the fact that Shamir secret-sharing scheme can be neither maximal or minimal. Another implication of it is that a scheme may be minimal if its share complexity is at least as high as that of DNF.
- Our second key result is a necessary condition for convertability to CNF from a broad class of (not necessarily linear) schemes. This result is proved via information-theoretic techniques and implies non-maximality for schemes with share complexity smaller than that of CNF.

We also provide a condition which is both necessary and sufficient for the existence of a share conversion to some linear scheme. The condition is stated as a system of linear equations, such that a conversion exists if and only if a solution to the linear system exists. We note that the impossibility results for linear schemes may be viewed as identifying a subset of contradicting equations in the system.

Another contribution of our paper, is in defining and studying share conversion for evolving secret-sharing schemes. In such a schemes, recently introduced by Komargodski et al. (IEEE ToIT'18), the number of parties is not bounded apriori, and every party receives a share as it arrives, which never changes in the sequel. Our impossibility results have implications to the evolving setting as well. Interestingly, unlike in the

standard setting, there is no maximum or minimum in a broad class of evolving schemes, even without any restriction on the share size.

Finally, we show that, generally, there is no conversion between additive schemes over different fields, even from CNF to DNF! However by relaxing from perfect to statistical security, it may be possible to convert, and exemplify this for (n, n) -threshold access structures.

1 Introduction

Secret-sharing is a fundamental notion in cryptography. A secret-sharing scheme enables a dealer to distribute a secret among a set of parties so that any pre-specified subset of qualified parties can recover the secret while any other subset of parties remains oblivious to the secret. The monotone class of subsets of qualified parties constitutes the *access structure* realized by the secret-sharing scheme.

Secret-sharing is a building block for realizing several complex cryptographic tasks. Certain such tasks may require additional properties in the secret-sharing scheme – for instance, succinctness of the shares, or homomorphism and other algebraic properties. This suggests use cases where a protocol requires secret-sharing according to one scheme during one stage, and according to another scheme during another. This motivated non-interactive conversion between secret-sharing schemes, which was formalized in [9] by Cramer, Damgård, and Ishai as *share conversion*.

We say there is a share conversion from a secret-sharing scheme Π to another scheme Π' implementing the same access structure Γ if each party can locally apply a deterministic function to their share to transform any valid secret-sharing under Π to a valid (but not necessarily random) secret-sharing of the same secret under Π' . In full generality, share conversion may be defined from Π to Π' which implement different access structures $\Gamma \supseteq \Gamma'$, respectively. Moreover the secret under Π' after the transformation can be a pre-specified function of the secret under Π before transformation. In this work, we focus on the natural case where $\Gamma = \Gamma'$ and the above-mentioned function is identity.

In the sequel, we will say $\Pi \geq \Pi'$ if there is a share conversion from Π to Π' . This induces a partial ordering over secret-sharing schemes realizing any access structure Γ . Many important insights into the partial order $\geq$ of convertability for *linear secret-sharing schemes* over a finite field were provided in [9]. Among other results, they proved that, for any access structure Γ and finite field $\mathbb{F}$, CNF-based secret-sharing scheme $CNF_{\Gamma, \mathbb{F}}$ is maximal, and DNF-based secret-sharing scheme $DNF_{\Gamma, \mathbb{F}}$ is minimal among the set of all linear secret-sharing schemes for Γ over $\mathbb{F}$. I.e., $CNF_{\Gamma, \mathbb{F}} \geq \Pi \geq DNF_{\Gamma, \mathbb{F}}$ for any linear secret-sharing scheme Π . Note, however, that the existence of additional minimal and maximal schemes is not ruled out in [9]. For certain access structures, specifically $(2,3)$ -threshold, they demonstrated that certain linear schemes like Shamir secret-sharing scheme is not maximal, as it is not convertible to CNF. They also show that a limited class of linear secret-sharing schemes – the so called *replicated schemes*, that are

similar in structure to CNF in the sense that the secret is defined as the sum of random elements, and every party gets a subset of them as it's share, are not maximal for (k, n) -threshold access structures unless they have share complexity as high as that of CNF (see Section 3.3 in [9]).

In this paper, we initiate a systematic study of convertibility between secret-sharing schemes, and obtain new results in several directions:

- We develop new and easily checkable necessary conditions for share conversion between linear schemes implementing a given access structure. These necessary conditions are in the form of linear algebraic constraints on the *monotone span program (MSP)* corresponding to the linear schemes. Using these conditions, we are able to get a clearer view of the partial order induced by convertibility over the linear schemes.
- We develop a necessary and sufficient condition for a conversion between the linear schemes Π, Π' in form of a linear system decided by the MSP of Π and Π' which has a solution if and only if $\Pi \geq \Pi'$.
- Next, we address the broader problem of share conversions involving potentially non-linear secret-sharing schemes. To this end, we introduce the notion of *non-degenerate* secret-sharing schemes and establish a necessary condition for converting shares from general (potentially non-linear) schemes to non-degenerate ones. This condition has implications for share conversion to CNF and Shamir secret-sharing schemes, which are both non-degenerate.
- We apply our results to develop necessary conditions for conversion to the well-studied schemes, such as CNF, DNF, and Shamir secret-sharing schemes.
- The necessary conditions we develop also bear consequences for secret-sharing schemes for evolving setting, i.e. where the number of parties is not bounded, and the party gets it's share when appears. We show that, for several interesting evolving access structures, there is no maximal or minimal scheme.
- We also initiate the study of share conversion between different fields. We show that, in a general case, there is no conversion between linear schemes over two different fields. To circumvent this, we propose a general approach of bounding the randomness domain in a source scheme. We build a leaky additive scheme over $\mathbb{Z}_p$ allowing conversion into $\mathbb{Z}_q$. As it's possible to see from our example, the proposed approach could result in a privacy leakage, which is often tolerable if small.

1.1 Overview of Our Results

In this section, we provide a brief exposition of our results, which we formally describe and prove in the subsequent sections.

Necessary conditions for conversion between linear schemes. A linear secret-sharing scheme Π over a field $\mathbb{F}$ implementing access structure Γ over n parties is characterized by a monotone span program described as a triple $(\mathbb{F}, M, \rho)$, where M is a matrix over $\mathbb{F}$ of dimension $m \times k$ and $\rho : [m] \rightarrow [n]$ defines the set of M 's rows corresponding to a certain party [15].

To share a secret $s \in \mathbb{F}$, the dealer samples a vector $\mathbf{r} \in \mathbb{F}^k$ such that its first coordinate is s , and computes $\mathbf{v} = M \cdot \mathbf{r}$. Then, the i 'th share in Π is $\text{sh}_i = \mathbf{v}[\rho^{-1}(i)]$, which is the sub-vector containing entries in the coordinates $\rho^{-1}(i)$. A qualified set of parties T can recover the secret using a reconstruction function $\alpha \in \mathbb{F}^{|\rho^{-1}(T)|}$ such that

$$(\alpha)^\top \cdot \mathbf{v}_T = (\alpha)^\top \cdot M_T \cdot \mathbf{r} = \mathbf{r}[1] = s.$$

Here, $\mathbf{v}_T = \mathbf{v}[\rho^{-1}(T)]$ and $M_T = M[\rho^{-1}(T), \cdot]$, i.e., the rows of M corresponding to the coordinates $\rho^{-1}(T)$ (under some pre-specified ordering).

One of the key tools in our paper is a necessary condition for conversion between a pair of linear schemes. Informally, it states that conversion is impossible, if the schemes satisfy certain linear-algebraic conditions.

Theorem 1 (Necessary condition for conversion between linear schemes - Informal). *There is no share conversion from a linear secret-sharing scheme described by $\text{MSP}(\mathbb{F}, M \in \mathbb{F}^{m \times k}, \rho)$ to another linear scheme $(\mathbb{F}, M' \in \mathbb{F}^{m' \times k'}, \rho')$ both realizing Γ , if there are sets of parties T, T^* and party $h \notin T \cup T^*$ such that $T \cup h \in \Gamma$, and no strict subsets of $T \cup h$ is qualified, and, when α and α' are reconstruction functions for $T \cup h$ in M and M' , respectively,*

1. $(\alpha'_h)^\top \cdot M'_h \in \text{Rowspan}(M'_{T^*})$.
2. $(\alpha_h)^\top \cdot M_h \notin (\text{Rowspan}(M_T) \cap \text{Rowspan}(M_h)) + (\text{Rowspan}(M_{T^*}) \cap \text{Rowspan}(M_h))$.

This theorem is formally stated as Theorem 11 in the technical section. We demonstrate the power of this seemingly abstract necessary condition by providing a concrete application to well studied secret-sharing schemes, and its application to share conversions for evolving access structures.

We exploit the fact that the share conversion function is local and the secret is preserved during the share conversion. We can reach a contradiction if it is possible to produce a pair of fooling instances of sharing under the source scheme that result in shares after conversion that do not respect the dependencies present among the shares in the target distribution. This proof is a vast generalization of the proof of a result in [9] that showed that in $(2, 3)$ -Shamir is not convertible to CNF, so that it applies to share conversion between any pair of linear schemes. Specifically, it follows that any Shamir secret-sharing scheme is not convertible to CNF.

In Theorem 12 we prove a different type of necessary condition for convertibility between linear schemes. As a corollary from Theorem 1 and Theorem 12, we prove that DNF is not convertible to any linear scheme with lower share complexity. In particular, it is not convertible to Shamir, which proves the non-minimality of the latter.

Both Theorem 12 and Theorem 1 are based on a key technical result – Lemma 1, which involves a single minterm T , while the theorems refer to at least 2 minterms to establish a conversion does not exist. We believe it is a useful conceptual simplification towards understanding convertability.

Convertibility characterization for linear schemes. We devise a characterization of convertibility between linear schemes by solvability of a certain system of linear equations $\mathcal{L}_{\Pi, \Pi'}$ which we provide in Section 5.

Theorem 2 (Theorem 14, informal). *There is a conversion from a linear scheme Π to a linear scheme Π' realizing the same access structure over the same field if and only if the linear system $\mathcal{L}_{\Pi, \Pi'}$ has a solution.*

A solution of the system encodes a conversion in a straightforward (although redundant) way. The high level idea is to solve for variables $X_{\mathbf{r}, i, j}$, where $X_{\mathbf{r}, i, j}$ represents the j 'th share of p_i , when converting from a sharing based on randomness $\mathbf{r}$ in Π (as is sometimes useful, here $\mathbf{r}$ is assumed to include s). We note that the impossibility in Theorem 1 may be viewed as identifying a subset of contradicting equations in the system, so that a solution does not exist.

Share conversion from non-linear schemes. In a prior work [9], CNF was proved to be maximal among linear schemes over the same field. In this work, we prove new necessary conditions for share conversion from general (potentially non-linear) schemes to CNF secret-sharing.

For this, we introduce the notion of *non-degenerate* secret-sharing schemes. A secret-sharing scheme Π is non-degenerate if any sub-scheme of Π is equivalent to Π . Here, a secret-sharing scheme Π' is a sub-scheme of Π if the secret domains of both schemes are identical, and every valid secret-sharing under Π' is also a valid secret-sharing under Π . A formal definition of non-degenerate secret-sharing schemes is provided in Definition 18.

To drive down this subtle notion, we will demonstrate that a 2-out-of-3 DNF secret-sharing scheme for secret domain $\{0, 1\}$ is *not* non-degenerate. In 2-out-of-3 DNF secret-sharing of a secret $s \in \{0, 1\}$, the shares of the 3 parties are $(r_{1,2}, s \oplus r_{3,1})$, $(s \oplus r_{1,2}, r_{2,3})$ and $(s \oplus r_{2,3}, r_{3,1})$, respectively, where $r_{1,2}$, $r_{2,3}$ and $r_{3,1}$ are uniform and independent bits. Consider a modified secret-sharing scheme in which the secret s is shared among the three parties exactly as in 2-out-of-3 DNF secret-sharing except that $r_{1,2}$, $r_{2,3}$ and $r_{3,1}$ are uniform and only pairwise independent bits. It is easy to see that the latter is a 2-out-of-3 secret-sharing which is also a sub-scheme of the former. However, they are not clearly not equivalent: the former uses more randomness; hence, 2-out-of-3 DNF is not non-degenerate.

On the other hand, CNF secret-sharing scheme for arbitrary access structures over a group $\mathbb{G}$ for secret domain $\mathbb{G}$, and Shamir secret-sharing scheme are non-degenerate as established in Theorem 15 and Theorem 16, respectively. Theorem 15 is proved by considering the correlation of the shares induced by picking a secret $s \in \mathbb{G}$ at random and secret-sharing it using any secret-sharing scheme Π that is a sub-scheme of the given CNF scheme. Appealing to correctness and privacy of Π , an information theoretic argument is used to show that the entropy of each share in this correlation is the same as that in the correlation obtained by secret-sharing a random secret s using the CNF scheme. Since the secret domain is the same as the group over which CNF is defined, this fur-

ther implies in a straightforward way that Π coincides with the CNF scheme, establishing its non-degeneracy.

By appealing to non-degeneracy of CNF, and using standard entropy lower bounds, we show that share conversion to CNF scheme is possible only if the share size under the source scheme is at least as large as that in the CNF scheme. The following result is formally stated in Theorem 17.

Theorem 3 (Extended maximality of CNF). *Let Π be a secret-sharing scheme realizing an n -party access structure Γ with secret domain $\mathbb{G}$ - a finite group. There is a share conversion from Π to CNF over $\mathbb{G}$ realizing Γ only if, for each $i \in [n]$, size of the share i in Π is at least $\log |\mathbb{G}| \cdot |\{F \in \mathcal{F} \text{ s.t. } i \notin F\}|$, where $\mathcal{F}$ is the set of all maximal forbidden sets associated with Γ .*

We note that Theorem 1 also implies results of non-maximality by impossibility of conversion to CNF for certain linear schemes Π with low share complexity. These results are mostly subsumed by Theorem 3, both because it does not restrict Π to be linear, and in terms of share size.

Share conversion for evolving secret-sharing. Komargodski et al. [16] defined evolving secret-sharing schemes where the unbounded number of parties arriving one after another, obtain their shares of secret. The previously qualified sets remain qualified, and shares of parties are not refreshed as new parties come, but each newcomer is provided a (potentially) progressively larger share. An evolving access structure is an infinite monotone class of qualified subsets of $\mathbb{N}$.

We initiate a study of share conversion for evolving secret-sharing, starting with formal extension of the notion of share conversion, MSP and linearity to the evolving setting. Then, we apply the theory we develop for proving impossibility of share conversion in the standard setting to the evolving setting. In particular, some of our results apply to *evolving linear secret-sharing schemes*, which have been previously considered in the literature, but never explicitly defined.

We address the problem of maximal and minimal secret-sharing schemes for evolving access structures, and show that for several broad classes of evolving schemes there is no maximal and minimal scheme. In Theorem 22 we formally state and prove the following result.

Theorem 4 (No evolving maximal scheme - Informal). *For any non-trivial evolving access structure, there exists no maximal secret-sharing scheme for one-bit secrets.*

By a non-trivial evolving access structure (See Definition 20), we mean one that does not devolve into a finite secret-sharing scheme among the first n parties (for some n) with the remaining parties either being not part of the qualified set or are required to simply receive the secret.

In the other direction, we obtain a slightly weaker result, showing there is no minimal linear scheme for certain access structures. This is formally stated and proved as Theorem 21.

Theorem 5 (No evolving minimal scheme - Informal). *For a certain broad class of evolving access structures Γ , and for the finite field $\mathbb{F}_2$, there is no minimal linear evolving scheme for any Γ in the class.*

Conversion between different fields. The bit simultaneously shared in two different fields, is called *dBit*, and is an important primitive for many applications, such as [2,6,8,11,12,18,19,23]. There exist bit share conversion protocols, here we point out only few of them, such as proposed in [6,7,10]. It is natural to raise the question if such a conversion can be done locally.

Generalizing our impossibility result for linear schemes over the same field, we prove the inconvertibility of the maximal *CNF* scheme over $\mathbb{Z}_p$ to any other linear scheme over $\mathbb{Z}_q$ for primes $p \neq q$ with the same secret domain $\{0, 1\}$.

We complement this result with a relaxed form of conversion between additive schemes for certain pairs of distinct primes for 1-bit shares. The relaxation allows for a privacy leakage (over the scheme randomness). It may be interesting to further explore share conversions with a small correctness errors. These may suffice for many applications, while potentially extending the set of convertible secret sharing scheme pairs (denote $\Pi \geq_\epsilon \Pi'$).

1.2 Future work

Our work leaves several fascinating questions open. The main question is to obtain a simpler characterization of convertibility between linear schemes. As a first step, identify pairs of linear schemes Π, Π' over the same field where Π is not convertible to Π' , which is not implied by Theorem 11 or Theorem 12. It may be particularly interesting to find a different type of conflicting requirements in the linear system in Section 5, thereby better understanding the easier linear case, which was also studied in the original paper on share conversion [9]. Another concrete question is to characterize the minimal and maximal schemes for various access structures (in other words, those convertible to CNF, or from DNF). As the linear systems introduced in Section 5 work also for non-linear source schemes, it could be also interesting to explore convertibility from such schemes to linear ones. This would require new techniques not based on theorems as above, that both rely on linearity of Π as well.

In evolving setting, proving impossibility results is potentially easier. In our context, it could be interesting to understand whether minimal and/or maximal schemes exist for access structures for which we have not resolved this question.

Finally, it is interesting to find new non-trivial examples of conversions which are possible. As an extension, it is interesting to study the direction of converting from a modified subset of a scheme Π where part of the randomness is removed, as we do for a modified version of additive over $\mathbb{Z}_p$ to $\mathbb{Z}_q$, and the incurred privacy losses. The motivation here is that some properties of the original Π may be preserved by such a transformation, which may suffice for certain applications.

2 Prior Work

Share conversion. Cramer et al. [9] first defined share conversion for secret-sharing schemes as a way for converting shares of a secret in one scheme into shares of the same secret in a different scheme using only local computation and no communication between parties. Referring to a conversion between schemes realizing the same access structure and defined over the same field, they showed that CNF can be converted to any linear scheme, and any linear scheme can be converted to DNF. Furthermore, they put forward an application of share conversion to improving efficiency of multiparty computation (MPC). Beimel et al. [5] use generalized share conversion including non-identity relation between secrets from $(2, 3)$ CNF to $(3, 3)$ additive secret-sharing over different groups to 3-party private information retrieval (PIR). In fact, they observe that certain share conversions are implicit in state-of-the-art 3-party PIR constructions from the literature, and devise another conversions along these lines that induces an improved PIR construction. They also put forward certain impossibility results for certain PIR induced conversions. The following papers [20,21] show additional positive results for potential conversions for 3-party schemes from the PIR-induced family.

Evolving secret-sharing. Komargodski et al. [16] defined evolving secret-sharing schemes for a case that the number of parties is unbounded, parties are only added as they arrive one after the other, and previously qualified sets remain qualified. They constructed the following evolving linear secret-sharing schemes: (1) a scheme for every evolving access structure, such that, the share size of the t^{th} party is 2^{t-1} ; (2) a k -threshold secret-sharing schemes in which the size of the share of party p_t is $O(k \log t)$; (3) an undirected st-connectivity schemes in which the share of each party is a bit.

A natural generalization of an evolving threshold access structure is to allow the threshold to depend on the index of the arriving party. Komargodski and Paskin-Cherniavsky [17] showed that any dynamic-threshold access can be realized with an evolving linear secret-sharing scheme in which the size of the share of party p_t is $O(t^4 \cdot \log t)$. Infinite decision trees were used in [16,17] to construct evolving secret-sharing schemes. Alon et al. [1] define formally this model. They showed how to construct evolving secret-sharing schemes for generalized infinite decision trees. We use this construction in our work.

Peter in [22] defined evolving conditional disclosure of secrets (CDS). In this model the number of parties is unbounded, parties arrive in sequential order. Each party holds a private input, and when arrives, it sends a random message to a referee. In turn, at any stage of the protocol, the referee should be able to reconstruct a secret string, held by all the parties, from the messages it gets, if and only if the inputs of the parties that arrived satisfy some condition.

3 Preliminaries

In this section, we present the necessary notation and formal definitions of secret-sharing schemes and evolving secret-sharing schemes.

Notation. For $n \in \mathbb{N}$ by $[n]$ we denote the set $\{1, 2, \dots, n\}$. We denote by $\log$ the logarithmic function with base 2. Vectors are denoted by bold letters (e.g., $\mathbf{r}$). For matrices M, M' with the same number of columns we denote by $[M; M']$ the concatenation of matrix M' below M . Similarly, for matrices M, M' with the same number of rows, $[M|M']$ is the concatenation of M' to the right for M . By $\text{Rowspan}(M)$ we denote the set of all vectors spanned by rows of M .

For a set of parties $\mathcal{P} = \{p_1, \dots, p_n\}$, when it is clear from the context, we often abuse notation replacing parties by their indexes from $[n]$. When we refer to a subset of parties $\{p_{i_1}, p_{i_2}, \dots, p_{i_t}\}$, we assume that $i_1 < i_2 < \dots < i_t$.

We will need the following well known fact in linear algebra.

Fact 6. *Let $\mathbb{F}$ be a field, and $A \in \mathbb{F}^{n \times k}$, $\mathbf{v} \in \mathbb{F}^{1 \times k}$ such that $\mathbf{v} \notin \text{Rowspan}(A)$. Then there exists a solution $\mathbf{r}$ to the linear system $[\mathbf{v}|A]\mathbf{r} = (1, 0, \dots, 0)$.*

Finally, we sometimes abuse notation, and use a linear subspace $L = \mathbb{F}^k$ as a matrix consisting of some basis of L as its rows, without explicitly stating so.

3.1 Secret-Sharing

We start by defining (perfect) secret-sharing schemes for a finite set of parties.

Definition 1 (Access Structures). *Let $\mathcal{P} = \{p_1, \dots, p_n\}$ be a set of parties. A collection $\Gamma \subseteq 2^{\{p_1, \dots, p_n\}}$ is monotone if $B \in \Gamma$ and $B \subseteq C$ imply that $C \in \Gamma$. An access structure $\Gamma \subseteq 2^{\{p_1, \dots, p_n\}}$ is a monotone collection of non-empty sets. Sets in Γ are called authorized, and sets not in Γ are called unauthorized. We will represent an n -party access structure by a function $f : \{0, 1\}^n \rightarrow \{0, 1\}$, where an input (i.e., a string) $\sigma = (\sigma_1, \sigma_2, \dots, \sigma_n) \in \{0, 1\}^n$ represents the set $A_\sigma = \{p_i : i \in [n], \sigma_i = 1\}$, and $f(\sigma) = 1$ if and only if $A \in \Gamma$. We will also call f an access structure.*

In a monotone access structure, the set $A \in \Gamma$ is called a minterm if there is no $B \subset A$ such that $B \in \Gamma$. The set $A \notin \Gamma$ is called a maxterm if for all $p_i \notin A$ it holds that $A \cup \{p_i\} \in \Gamma$.

The most basic and well-known access structure is the threshold access structure:

Definition 2 (Threshold Access Structures). *Let $1 \leq k \leq n$. A k -out-of- n threshold access structure Γ over a set of parties $\mathcal{P} = \{p_1, \dots, p_n\}$ is the access structure containing all subsets of size at least k , that is, $\Gamma = \{A \subseteq \mathcal{P} : |A| \geq k\}$.*

A secret-sharing scheme defines a way to distribute shares to parties. Such a scheme is said to realize an access structure Γ if the shares held by any authorized set of parties (i.e., a set in the access structure) can be used to reconstruct the secret, and the shares held by any unauthorized set of parties reveal nothing about the secret. The formal definition is given as follows.

Definition 3 (Secret-Sharing Schemes). A secret-sharing scheme Π over a set of parties $\mathcal{P} = \{p_1, \dots, p_n\}$ with domain of secrets S and domain of random strings R is a mapping from $S \times R$ to a set of n -tuples $S_1 \times S_2 \times \dots \times S_n$ (the set S_j is called the domain of shares of p_j). A dealer distributes a secret $s \in S$ according to Π by first sampling a random string $r \in R$ with uniform distribution, computing a vector of shares $\Pi(s; r) = (\text{sh}_1, \dots, \text{sh}_n)$, and privately communicating each share sh_j to party p_j . For a set $A \subseteq \{p_1, \dots, p_n\}$, we denote $\Pi_A(s; r)$ as the restriction of $\Pi(s; r)$ to its A -entries (i.e., the shares of the parties in A).

A secret-sharing scheme Π with domain of secrets S realizes an access structure Γ if the following two requirements hold:

Correctness. The secret s can be reconstructed by any authorized set of parties.

That is, for any authorized set $B = \{p_{i_1}, \dots, p_{i_{|B|}}\} \in \Gamma$, there exists a reconstruction function $\text{Recon}_B : S_{i_1} \times \dots \times S_{i_{|B|}} \rightarrow S$ such that for every secret $s \in S$ and every random string $r \in R$, it holds that $\text{Recon}_B(\Pi_B(s; r)) = s$.

Security. Every unauthorized set cannot learn anything about the secret from its shares. Formally, for any set $T \notin \Gamma$, every two secrets $s_1, s_2 \in S$, and every possible vector of shares $\langle \text{sh}_j \rangle_{p_j \in T}$,

$$\Pr \left[\Pi_T(s_1; r) = \langle \text{sh}_j \rangle_{p_j \in T} \right] = \Pr \left[\Pi_T(s_2; r) = \langle \text{sh}_j \rangle_{p_j \in T} \right],$$

where the probability is over the choice of r from R with uniform distribution.

The size of the share of party p_j is defined as $\log |S_j|$ and the size of the shares of Π as $\max_{1 \leq j \leq n} \log |S_j|$. The total share size of Π is defined as $\sum_{j=1}^n \log |S_j|$.

Next we give some widely known secret-sharing schemes.

Definition 4 (Additive Secret-Sharing Scheme [14]). In the additive secret-sharing scheme $\text{ADD}_{\mathbb{F}, n}$ over $\mathbb{F}$, shares $\text{sh}_1, \dots, \text{sh}_n$ are sampled uniformly at random from $\mathbb{F}$ on the condition that $s = \sum_{i=1}^n \text{sh}_i$, and $\Gamma = \{\mathcal{P}\}$.

Definition 5 (Shamir Secret-Sharing Scheme [24]). In the (n, k) -Shamir secret-sharing scheme over $\mathbb{F}$ realizing k -out-of- n threshold access structure Γ , the dealer sets a polynomial $p(x) = s + r_1x + \dots + r_{k-1}x^{k-1}$ by uniformly random sampling of $r_j \leftarrow \mathbb{F}$ for $j \in [k-1]$. The share of p_i for $i \in [n]$ is set as $\text{sh}_i = p(i)$.

The properties of Shamir's scheme over $\mathbb{F}_{2^m}$ for an appropriate $m \in \mathbb{N}$ are summarized in the next theorem.

Theorem 7 (Shamir [24]). For every $n \in \mathbb{N}$, and $k \in [n]$, there is a secret-sharing scheme for secrets of size ℓ (i.e., the domain of secrets is $S = \{0, 1\}^\ell$) realizing the k -out-of- n threshold access structure, in which the share size is $\max\{\ell, \lceil \log(n+1) \rceil\}$. Moreover, the shares of the scheme are elements of the field $\mathbb{F}_{2^{\ell + \log n}}$.

Next two schemes realize any monotone access structure. A replicated secret-sharing scheme [13] is also known as a CNF secret-sharing scheme [14].

Definition 6 (Replicated Secret-Sharing Schemes [13]). Let $\Gamma \subseteq 2^{[n]}$ be a (monotone) access structure, and let $\mathcal{T}$ is the set of all maxterms of Γ . The CNF secret-sharing schemes for Γ over $\mathbb{F}$, denoted $CNF_{\Gamma, \mathbb{F}}$, proceeds as follows. A secret $s \in S$ is shared in $ADD_{\mathbb{F}, |\mathcal{T}|}$, where each share r_T is labelled by a different set $T \in \mathcal{T}$. Then, the dealer distributes to each party p_j all shares r_T such that $j \notin T$, that is, $\text{sh}_j = (r_T)_{j \notin T}$. For correctness, since Γ is monotone, a qualified set $Q \in \Gamma$ cannot be contained in any unqualified set, hence, members of Q jointly view all shares r_T and can thus reconstruct the secret s . For privacy, the parties of every maxterm $T \in \mathcal{T}$ jointly miss exactly one additive share r_T , hence parties of any unqualified set miss at least one share.

Definition 7 (DNF Secret-Sharing Scheme [14]). In DNF secret-sharing schemes, denoted $DNF_{\Gamma, \mathbb{F}}$, the secret s is additively shared between the parties of each minterm, where each additive sharing uses independent randomness.

More secret-sharing schemes can be defined using the notion of a monotone span program (MSP). We bring the definition of MSP below.

Definition 8 (Monotone Span Program [15]). A monotone span program is a triple $\mathcal{M} = (\mathbb{F}, M, \rho)$, where $\mathbb{F}$ is a field, M is an $m \times k$ matrix over $\mathbb{F}$, and a mapping $\rho : [m] \rightarrow [n]$ labels each row of M by a party's index. The size of $\mathcal{M}$ is the number of rows of M (i.e., m).

Next we give some notation which simplifies addressing to sets and operations of MSP. Let $M \in \mathbb{F}^{m \times k}$ be a matrix, and $A \subseteq [m]$. We denote by $M[A, \cdot]$ the $|A| \times k$ dimensional submatrix that restricts M to the rows labelled by $i \in A$. Hence, for an MSP $(\mathbb{F}, M \in \mathbb{F}^{m \times k}, \rho)$ describing an n -party linear secret-sharing scheme, and $h \in [n]$, $M[\rho^{-1}(h), \cdot]$ denotes the submatrix induced by rows of M corresponding to shares of party h . For any $S \subseteq [n]$, for brevity, we will refer to $M[\rho^{-1}(S), \cdot]$ by M_S , and when $S = \{h\}$ for some $h \in [n]$, we will further simplify the notation by referring to $M_{\{h\}}$ as M_h . Similarly, in the context of the above MSP, for a column vector $\alpha \in \mathbb{F}^m$, and set $A \subseteq [m]$, we denote by $\alpha[A]$ the sub-vector of α labeled by $i \in A$, and for the subset of parties $S \subseteq [n]$ we let $\alpha_S = \alpha[\rho^{-1}(S)]$, and $\alpha_{\{h\}} = \alpha_h$.

Definition 9 (Access structure accepted by MSP [15]). We say that MSP $\mathcal{M}$ accepts $B \subseteq [n]$ if the rows of M_B span the vector $\mathbf{e}_1 = (1, 0, \dots, 0)$, called a target vector.³ We say that $\mathcal{M}$ accepts an access structure Γ if $\mathcal{M}$ accepts a set B if and only if $B \in \Gamma$.

A monotone span program implies a so called *linear secret-sharing scheme* for an access structure containing all the sets accepted by the program. Essentially, a dealer gives each party the rows of matrix M assigned to it, multiplied by the randomness vector.

³ In [15] it is proven that one could define MPS's with any target vector $\epsilon \neq \mathbf{0}$, rather than $\mathbf{e}_1$, resulting in the same matrix size and labeling.

Claim 1 ([4]). Let $\mathcal{M} = (\mathbb{F}, M, \rho)$ be a MSP accepting an access structure Γ , where $\mathbb{F}$ is a finite field and for every $j \in [n]$ there are a_j rows of M labeled by p_j . Then, there is a linear secret-sharing scheme realizing Γ for $S = \mathbb{F}$ such that the share of party p_j is a vector in $\mathbb{F}^{a_j}$ with the information equal to $\max_{1 \leq j \leq n} a_j$.

3.2 Evolving Secret-Sharing Schemes

In an evolving secret-sharing scheme, defined by [16], the number of parties is unbounded. Parties arrive one after the other; when a party p_t arrives the dealer gives it a share. The dealer cannot update the share later and does not know how many parties will arrive after party p_t . Thus, we measure the share size of p_t as a function of t . We start by defining an evolving access structure, which specifies the authorized sets. The number of parties in an evolving access structure is infinite, however every authorized set is finite.

Definition 10 (Evolving Access Structures). Let $\mathcal{P} = \{p_i\}_{i \in \mathbb{N}}$ be an infinite set of parties. A collection of finite sets $\Gamma \subseteq 2^{\mathcal{P}}$ is an evolving access structure if for every $t \in \mathbb{N}$ the collections $\Gamma^t \triangleq \Gamma \cap 2^{\{p_1, \dots, p_t\}}$ is an access structure as defined in definition 1. We will represent an access structure by a function $f : \{0, 1\}^* \rightarrow \{0, 1\}$, where an input (i.e., a string) $\sigma = (\sigma_1, \sigma_2, \dots, \sigma_n) \in \{0, 1\}^n$ represents the set $A_\sigma = \{p_i : i \in [n], \sigma_i = 1\}$,⁴ and $f(\sigma) = 1$ if and only if $A_\sigma \in \Gamma$. We will also call f an evolving access structure.

Definition 11 (Evolving Secret-Sharing Schemes). Let S be a domain of secrets, where $|S| \geq 2$, and $\{R_t\}_{t \in \mathbb{N}}, \{S_t\}_{t \in \mathbb{N}}$ be two sequences of finite sets. An evolving secret-sharing scheme with domain of secrets S is a sequence of mappings $\Pi = \{\Pi^t\}_{t \in \mathbb{N}}$, where for every $t \in \mathbb{N}$, Π^t is a mapping $\Pi^t : S \times R_1 \times \dots \times R_t \rightarrow S_t$ (this mapping returns the share sh_t of p_t).

An evolving secret-sharing scheme $\Pi = \{\Pi^t\}_{t \in \mathbb{N}}$ realizes an evolving access structure Γ if for every $t \in \mathbb{N}$ the secret-sharing scheme $\Pi_t(s; r_1, \dots, r_t) \triangleq \langle \Pi^1(s; r_1), \dots, \Pi^t(s; r_1, \dots, r_t) \rangle$ (i.e., the shares of the first t parties) is a secret-sharing scheme realizing Γ^t according to definition 3.

By default, the domain of secrets of an evolving secret-sharing scheme is $\{0, 1\}$. Known results show that every evolving access structure can be realized by an evolving secret-sharing scheme.

Definition 12 (Evolving Threshold Access Structures). Let $k \in \mathbb{N}$. The evolving k -threshold access structure is the evolving access structure Γ , where Γ^t is the k -out-of- t threshold access structure.

Komargodski et al. [16] showed that any evolving threshold access structure can be realized by an efficient evolving secret-sharing scheme.

⁴ In particular, the same set has infinitely many representations by inputs of various lengths, using sufficiently many trailing zeros.

Theorem 8 ([16]). *For every $k \in \mathbb{N}$, there is a secret-sharing scheme realizing the evolving k -threshold access structure such that the share size of party p_t is $(k - 1) \cdot \log t + \text{poly}(k) \cdot o(\log t)$.*

Definition 13 (Evolving Undirected st-connectivity Access Structures).

An evolving undirected st-connectivity access structure is defined as follows. The parties in the access structure are the edges of an undirected graph $G = (V, E)$, where V is countably infinite, with some order on the edges that specifies the order that the parties arrive. There are two fixed vertices in the graph $u_s, u_t \in V$, where u_s is called the source vertex and u_t the target vertex. A finite set of parties (i.e., edges) is authorized if and only if they contain an undirected path from u_s to u_t .

Komargodski et al. [16] showed that every undirected st-connectivity access structure can be realized by an evolving secret-sharing scheme in which the share of each party is a bit.

The next two structures, infinite decision trees and generalized decision trees, are used to define more general evolving secret sharing access structures.

Definition 14 (Generalized Infinite Decision Trees – GIDT, [1]). *A generalized infinite decision tree is a quadruple $T = (G = (V, E), u_0, \mu, h)$, where*

- V is a countable set of vertices.
- $G = (V, E)$ is an infinite directed tree with root vertex u_0 such that the out-degree of each vertex is finite. We denote that i^{th} level L_i as $\{u \in V : u \text{ is at distance } i \text{ from } u_0\}$, and refer to L_i as the i^{th} layer.
- $h : \mathbb{N} \rightarrow \mathbb{N}$ is an increasing function that partitions the variables into generations, where for $i \in \mathbb{N}$, generation i is the variables $G_i \triangleq \{x_{h(i-1)+1}, \dots, x_{h(i)}\}$ (where we define $h(0) = 0$).
- μ is a labeling of the edges by predicates, where for every edge e from level L_{i-1} to L_i , the labeling μ_e is some monotone predicate on the variables of generation i , of the form $\varphi(x_{h(i-1)+1}, \dots, x_{h(i)}) : \{0, 1\}^{h(i)-h(i-1)} \rightarrow \{0, 1\}$.

For a path P in the tree ending at a vertex in layer i , we say that P is satisfied by an input $\sigma \in \{0, 1\}^t$, denoted by $\text{sat}_P(\sigma) = 1$, if $h(i) \leq t$ (that is, the variables in all predicates labeling edges in P are from $x_1, \dots, x_t$) and for each edge e on the path the predicate μ_e is satisfied by σ . The GIDT T accepts an input σ if there is at least one directed path P starting in the source vertex u_0 and leading to a leaf such that $\text{sat}_P(\sigma) = 1$ (where $\text{sat}_P(\sigma) = 1$ if σ satisfies all variables on the path). The function $f : \{0, 1\}^ \rightarrow \{0, 1\}$ computed by T is the function f such that $f(\sigma) = 1$ if and only if T accepts σ .*

Definition 15 (Infinite decision trees – IDT). *An infinite decision tree $T = (G = (V, E), u_0 = 0, \mu)$ is a special case of GIDT, where each edge (u, v) is either labeled by the constant 1 or by a variable x_v , where for simplicity we assume that $V = \mathbb{N} \cup \{0\}$ (i.e., a vertex is a non-negative integer). As G is a tree, each variable labels at most one edge. Furthermore, we assume that the vertices are ordered by the layers, i.e., $L_0 = \{0\}$, $L_1 = \{1, \dots, w(1)\}$, and so on (where*

$w(i)$ is the width of layer L_i). The variables in generation i are $\{x_j : j \in L_i\}$ (thus, we do not need to specify h for an IDT).

Infinite decision trees were used in [16,17] to construct evolving secret-sharing schemes. Alon et al. [1] showed how to construct secret-sharing schemes for IDT.

Construction 9 (An Evolving Secret-Sharing Scheme Π_{IDT} for an IDT

$T = (G, u_0, \mu)$).

Input: $s \in \{0, 1\}$.

The sharing algorithm:

- For $i = 1$ to ∞ :
 - For every vertex $u \in L_{i-1}$ and $v \in L_i$, when party p_v arrives choose a bit r_v as follows:
 - * If v is a leaf, then let $u_0, v_1, \dots, v_{t-1}, v$ be the path from the root u_0 to v in G and assign $r_v \leftarrow s \oplus \bigoplus_{j=1}^{t-1} r_{v_j}$.
 - * If v is not a leaf and $\mu_{(u,v)} = x_v$, then r_v is a uniformly distributed random bit.
 - * If v is not a leaf and $\mu_{(u,v)} = 1$, then $r_v \leftarrow 0$.
 - The share of p_v is $\text{sh}_v = r_v$.

Claim 2 ([1]). The evolving secret-sharing scheme Π_{IDT} realizes the infinite decision tree $T = (G, u_0, \mu)$, where the share of p_t is a bit.

Alon et al. [1] defined generalized infinite decision trees and showed how to construct secret-sharing schemes for GIDT.

Construction 10 (An Evolving Secret-Sharing Scheme Π_{GIDT} for a GIDT

$T = (G = (V, E), u_0, \mu, h)$).

Input: $s \in \{0, 1\}$.

- Construct from the GIDT $T = (G = (V, E), u_0, \mu, h)$ an IDT $T' = (G = (V, E), u_0, \mu')$ whose variables are $\{y_i : i \in \mathbb{N}\}$, where for every edge $(u, v) \in E$ if the predicate $\mu_{(u,v)}$ is the constant predicate 1, then $\mu'(u, v) = 1$; otherwise $\mu'(u, v) = y_v$.
- Execute the scheme Π_{IDT} for T' and use its shares as follows:
 - (* Recall that in Π_{IDT} the parties arrive according to layers, where inside a layer the order is some arbitrary fixed order *)
- For $i = 1$ to ∞ do:
 - When party $p_{h(i-1)+1}$ arrives do:
 - * For every $(u, v) \in E$, where $u \in L_{i-1}$, $v \in L_i$, and $\mu_{(u,v)} \neq 1$, generate the share r_v of y_v in the scheme Π_{IDT} and share r_v using a secret-sharing scheme realizing the access structure defined by $\mu_{(u,v)}$ among the parties $p_{h(i-1)+1}, \dots, p_{h(i)}$.
 - * Let sh_t , for $h(i-1)+1 \leq t \leq h(i)$, be the concatenation of the shares of p_t in all these schemes.
 - * Give party $p_{h(i-1)+1}$ the share $\text{sh}_{h(i-1)+1}$.
 - For $t = h(i-1)+2$ to $h(i)$ do:
 - * When party p_t arrives give it the share sh_t .

Proposition 1 ([1]). There exists an evolving secret-sharing scheme that realizes the GIDT $T = (G, u_0, \mu, h)$.

3.3 Share Conversion

Cramer et al. [9] defined the notion of a share conversion as a local mapping from the shares a secret over one scheme into shares over another scheme, maintaining the secret value. We next include a formal definition of share conversion.⁵

Definition 16 (Share Conversion). *Let Π, Π' be two secret-sharing schemes over the same secret-domain S for n parties realizing the same access structure. We say that Π is locally convertible to Π' if there exist functions $g_1, \dots, g_n$ such that the following holds. If $(\text{sh}_1, \dots, \text{sh}_n)$ are valid shares of a secret s in Π (i.e., $\Pr[\Pi(s; \mathbf{r}) = (\text{sh}_1, \dots, \text{sh}_n)] > 0$), then $(g_1(\text{sh}_1), \dots, g_n(\text{sh}_n))$ are valid shares of the same secret s in Π' . We denote by g the concatenation of all g_i , namely $g(\text{sh}_1, \dots, \text{sh}_n) = (g_1(\text{sh}_1), \dots, g_n(\text{sh}_n))$, and refer to g as a conversion function.*

We next extend the definition of share conversion to the evolving setting.

Definition 17 (Evolving Share Conversion). *Let Π, Π' be two evolving secret-sharing schemes over the same secret-domain S realizing an access structure Γ . We say that Π is locally convertible to Π' if there exists a sequence of functions $g_1, g_2, g_3, \dots$ such that the following holds. For every $t \geq 1$, if $(\text{sh}_1, \dots, \text{sh}_t)$ are valid shares of a secret s in Π (i.e., $\exists \mathbf{r} \in R_1 \times \dots \times R_t$ such that $\Pi(s; \mathbf{r}) = (\text{sh}_1, \dots, \text{sh}_t)$), then $(g_1(\text{sh}_1), \dots, g_t(\text{sh}_t))$ are valid shares of the same secret s in Π' . We denote by g the concatenation of all g_i , namely $g(\text{sh}_1, \text{sh}_2, \dots) = (g_1(\text{sh}_1), g_2(\text{sh}_2), \dots)$, and refer to g as a conversion function.*

If the secret-sharing scheme Π is convertible to Π' , we say that $\Pi \geq \Pi'$. This defines a partial ordering over secret-sharing schemes.

Next, we show that changing a target vector preserves much of the MSP structure, while being convertible to the original scheme.

Claim 3. Let $\Pi = (\mathbb{F}, M \in \mathbb{F}^{m \times k}, \rho)$ is a linear scheme for an access structure Γ with the target vector $\epsilon \in \mathbb{F}^m \setminus \mathbf{0}$. Then for any target vector $\epsilon' \in \mathbb{F}^m \setminus \mathbf{0}$, there exists a linear scheme $\Pi' = (\mathbb{F}, M' \in \mathbb{F}^{m \times k}, \rho)$ for Γ , convertible to Π .

Proof. We prove that the identity conversion works. Let H denote an invertible matrix such that $\epsilon \cdot H = \epsilon'$. Let $\Pi' = (\mathbb{F}, M' = M \cdot H, \rho)$ with target vector ϵ' . First, observe that Π' is indeed a scheme implementing Γ . This is the case as the submatrix $M \cdot H[A, \cdot]$ spans ϵ' if and only if $M[A, \cdot]$ spans ϵ (and the fact we keep ρ the same). It is known [4] that the reconstructing sets A are exactly those that span the target vector.

To see that Π' is convertible to Π via the identity transformation g , we observe that for every $\mathbf{r} \in \mathbb{F}^m$, we have $M' \cdot \mathbf{r} = M \cdot (H \cdot \mathbf{r})$, that is, it corresponds to a valid sharing via randomness $H\mathbf{r}$. Now show that the shared secret value $\langle \epsilon', \mathbf{r} \rangle$ is not changed, i.e., $\langle \epsilon, H\mathbf{r} \rangle = \langle \epsilon', \mathbf{r} \rangle$. Let α denote a reconstruction vector for M . We have

$$(\alpha)^\top \cdot M(H \cdot \mathbf{r}) = (\epsilon)^\top \cdot H \cdot \mathbf{r} = \langle (\epsilon)^\top H, \mathbf{r} \rangle = \langle (\epsilon')^\top, \mathbf{r} \rangle. \quad \square$$

⁵ In [9], they in fact give a slightly more general definition.

In linear (MSP-based) schemes, it is convenient to consider a secret s as part of the randomness vector $\mathbf{r}$, being its first coordinate. Sometimes, s is defined by $\mathbf{r}$ in a different manner, which results in a different than $\mathbf{e}_1$ target vector in MSP. For example, in CNF with $\mathbf{1}$ target, as used in [9], the secret is the sum of all elements in $\mathbf{r}$. Thus, we will sometimes consider conversions to a scheme Π' with a certain target vector, and implicitly rely on the implied conversion to Π' with a different target vector.

4 Impossibility results for linear Share Conversion

Our impossibility results for linear schemes presented in this section follow from the lemma stated below.

Lemma 1. *Let Π, Π' denote linear secret-sharing schemes realizing Γ and specified by MSPs $(\mathbb{F}, M \in \mathbb{F}^{m \times k}, \rho), (\mathbb{F}, M' \in \mathbb{F}^{m' \times k'}, \rho')$. Let $T \cup \{h\}$ denote a minterm in Γ for $T \subseteq [n], h \in [n]$, and let $\alpha_{T \cup h}, \alpha'_{T \cup h}$ be its reconstruction functions for Π and Π' , respectively. Let $L = \text{Rowspan}(M_T) \cap \text{Rowspan}(M_h)$, and B denote a basis for L . Suppose g is some share conversion from Π to Π' . Then $(\alpha'_h)^\top g_h(M_h \mathbf{r}) = (\alpha_h)^\top M_h \mathbf{r} + c(B\mathbf{r})$ for some function c , where $\mathbf{r}$ is a randomness vector for Π .⁶*

Remark 1. The lemma gives us a high level clue as to why CNF is high up in the convertibility (partial) order for linear schemes of a given access structure, while DNF is so low. In DNF, for every party h , and every minterm $T \cup \{h\}$, it is easy to see that the resulting $L = \{\mathbf{0}\}$. This way, there is “little freedom” in converting determining the converted (sub)share for h (this is formally proved in Theorem 13). On the other hand, in CNF, for a given h and minterm $T \cup \{h\}$, $\dim(L)$ is typically large - as large as the number of maxterms H where $H \cap (T \cup \{h\}) \subset T$. This allows for more freedom at choosing the conversion function.

Proof. (Of Lemma 1) First observe that since $T \cup \{h\}$ is a minterm, $(\alpha_h)^\top M_h \notin \text{Rowspan}(M_T)$, and hence $(\alpha_h)^\top M_h \notin L$. Also, by definition, $B \subseteq \text{Rowspan}(M_h)$. Note that $\{(\alpha_h)^\top M_h\} \cup B$ may not constitute a basis of the rows of M_h , in which case, it is complemented by adding a set of appropriate linear combinations of an appropriate set X of rows in M_h . Let M_T^- denote a subset of M_T 's rows constituting a basis of $\text{Rowspan}(M_T)$. By choice of B , for any scalar a and vector $\mathbf{v}$ (of the right dimension) there exists randomness $\mathbf{r}$ (one or more) such that $M_T^- \mathbf{r} = \mathbf{v}$, $(\alpha_h)^\top M_h \mathbf{r} = a$. Note that $B\mathbf{r}$ is determined by $\mathbf{v}$ (and is otherwise independent of $\mathbf{r}$). In the sequel, for (an implicit or explicit) randomness vector $\mathbf{r}$, we let $\mathbf{v}, a$ denote the share portions as above induced by it. As α' is a reconstruction function, and by locality of g and linearity of Π' , $\forall \mathbf{r}$ it holds that

$$(\alpha'_{T \cup h})^\top g_{T \cup h}(M_{T \cup h} \mathbf{r}) = (\alpha'_h)^\top g_h(M_h \mathbf{r}) + (\alpha'_T)^\top g_T(\mathbf{v}) = s \quad (1)$$

⁶ Note that even if $L = \{\mathbf{0}\}$, we are free to pick the constant $c(\mathbf{0})$, which depends only on α in this case.

By Equation (1), and the definition of α we have that

$$\begin{aligned} (\alpha'_h)^\top g_h(M_h \mathbf{r}) &= s - (\alpha'_T)^\top g_T(M_T \mathbf{r}) \\ &= (\alpha_h)^\top M_h \mathbf{r} + (\alpha_T)^\top M_T \mathbf{r} - (\alpha'_T)^\top g_T(M_T \mathbf{r}), \end{aligned} \quad (2)$$

where the first term equals to a by definition, and the rest is some function of only $\mathbf{v}$. We denote $(\alpha_T)^\top M_T \mathbf{r} - (\alpha'_T)^\top g_T(M_T \mathbf{r})$ by $f(\mathbf{v})$. From locality of g , $f(\mathbf{v})$ in fact depends only on $B\mathbf{r}$. To see this, denote $\text{span}(M_T^-) = \text{span}(B) \oplus \text{span}(B_T^-)$ (direct sum of linear subspaces) for an independent set of vectors $B_T^- \subseteq \text{Rowspan}(M_T)$ complementing B into a basis. Using the fact that B is a basis of $\text{Rowspan}(M_h) \cap \text{Rowspan}(M_T)$, it is easy to show that B_T^- with $\text{span}(B_T^-) \cap \text{Rowspan}(M_h) = \{\mathbf{0}\}$ indeed exists.

The observation follows by locality, since g_h has no information on $\text{span}(B_T^-)\mathbf{r}$ (given $B\mathbf{r}$ that it knows). Also, $(\alpha'_h)^\top g_h(M_h \mathbf{r})$ may not depend on $X\mathbf{r}$, as $\text{span}(X) \cap \text{Rowspan}(M_T) = \{\mathbf{0}\}$, as then it would not be a (deterministic) function of $\mathbf{v}$.⁷ Thus, we have

$$(\alpha'_h)^\top g_h(M_h \mathbf{r}) = a + c(B\mathbf{r}) \quad (3)$$

for some function c , as required. $\square$

Next we state two corollaries defining necessary conditions for convertability between linear schemes Π, Π' (as sufficient conditions for non-convertability).

The following theorem is in a sense a generalization of Claim in Section 3.3 in [9] of non-convertability from Shamir to CNF for $(2, 3)$ -threshold access structures.

Theorem 11. *Let Γ be an access structure on n parties. Let Π and Π' be linear secret-sharing schemes realizing Γ and specified by MSP $(\mathbb{F}, M \in \mathbb{F}^{m \times k}, \rho)$ and $(\mathbb{F}, M' \in \mathbb{F}^{m' \times k'}, \rho')$, respectively. Then, Π has no share conversion to Π' if there exist $h \in [n]$, $\emptyset \neq T \subseteq [n] \setminus \{h\}$ such that $T \cup \{h\}$ is a minterm of Γ with the reconstruction functions α and α' in Π and Π' resp., and $\emptyset \neq T^* \subseteq [n] \setminus \{h\}$ that satisfy the following conditions:*

1. $(\alpha'_h)^\top \cdot M'_h \in \text{Rowspan}(M'_{T^*})$.
2. $(\alpha_h)^\top \cdot M_h \notin (\text{Rowspan}(M_T) \cap \text{Rowspan}(M_h)) + (\text{Rowspan}(M_{T^*}) \cap \text{Rowspan}(M_h))$.⁸

⁷ Note that the above does not imply that $g_h(M_h \mathbf{r})$ does not depend on $X\mathbf{r}$, but rather that it is of the form $g'_h(B\mathbf{r}, a) + g''_h(X\mathbf{r})$, where $g''_h(X\mathbf{r}) = (g''_{h,1}(X\mathbf{r}), \dots, g''_{h,\ell}(X\mathbf{r}))$ is a vector of functions in the formal variables $X[1]\mathbf{r}, \dots, X[\ell]\mathbf{r}$, such that $(\alpha'_h)^\top g''_h(X\mathbf{r}) = 0$. Similarly, it implicitly follows that the vector $g_T(M_T \mathbf{r}) = g_T(B\mathbf{r}, B_T^- \mathbf{r})$, of functions is such that $(\alpha'_T)^\top g''_T(B_T^- \mathbf{r})$ is a formal function of $B\mathbf{r}$ only. That is in Fourier basis in the variables $y_1 = B[1]\mathbf{r}, \dots, y_{|B|} = B[|B|]\mathbf{r}, \dots, y_i = B_T^-[i-|B|]\mathbf{r}, \dots, y_k$, it does not contain minterms of the form $\prod_{j \in A} y_j$ where $A \setminus [|B|]$ is not empty. This also poses a certain restriction of g_T which can possibly be exploited by future work to get closer to a simple characterization of convertibility.

⁸ Recall that $U + V = \{u + v | u \in U, v \in V\}$.

Proof. Let us assume for contradiction a conversion g exists, and denote $L = \text{Rowspan}(M_T) \cap \text{Rowspan}(M_h)$ and $L^* = \text{Rowspan}(M_{T^*}) \cap \text{Rowspan}(M_h)$. We first show that there exist randomness vectors $\mathbf{r}_1, \mathbf{r}_2$ such that:

1. $(L + \text{Rowspan}(M_{T^*}))\mathbf{r}_1 = (L + \text{Rowspan}(M_{T^*}))\mathbf{r}_2 = \mathbf{0}$.
2. $(\alpha_h)^\top M_h \mathbf{r}_1 \neq (\alpha_h)^\top M_h \mathbf{r}_2$.

Fix $\mathbf{r}_1 = \mathbf{0}$. Then, $(L + \text{Rowspan}(M_{T^*}))\mathbf{r}_1 = \mathbf{0}$ and $(\alpha_h)^\top M_h \mathbf{r}_1 = 0$. Next, let us show that $\mathbf{r}_2$ as required exists. We claim

$$\text{Rowspan}(M_h) \cap (L + L^*) = \text{Rowspan}(M_h) \cap (L + \text{Rowspan}(M_{T^*})).$$

Clearly, former is contained in the latter. For the other direction, consider $x \in \text{Rowspan}(M_h) \cap (L + \text{Rowspan}(M_{T^*}))$. Since L is a subspace of M_h , it suffices to show that if $x \in L + y$ such that $y \in \text{Rowspan}(M_{T^*})$, then $y \in \text{Rowspan}(M_{T^*}) \cap \text{Rowspan}(M_h) = L^*$. Indeed, if $y \in \text{Rowspan}(M_{T^*}) \setminus \text{Rowspan}(M_h)$, then $y + z \notin \text{Rowspan}(M_h)$ for all $z \in L$, which contradicts the fact $x \in \text{Rowspan}(M_h)$; the claim follows.

By this claim and condition 2 of the Theorem, $(\alpha_h)^\top M_h \notin L + \text{Rowspan}(M_{T^*})$ since $(\alpha_h)^\top M_h \in \text{Rowspan}(M_h)$. Therefore, there exists $\mathbf{r}_2$ such that $(L + \text{Rowspan}(M_{T^*}))\mathbf{r}_2 = \mathbf{0}$ and $(\alpha_h)^\top M_h \mathbf{r}_2 = 1 (\neq 0)$, by Fact 6.

By Lemma 1 applied to h, T , when B is the basis for L , we have

$$(\alpha'_h)^\top g_h(M_h \mathbf{r}_1) \neq (\alpha'_h)^\top g_h(M_h \mathbf{r}_2), \quad (4)$$

since $B\mathbf{r}_1 = B\mathbf{r}_2 (= \mathbf{0})$ and $(\alpha_h)^\top M_h \mathbf{r}_1 \neq (\alpha_h)^\top M_h \mathbf{r}_2$. On the other hand, by locality, $g_{T^*}(M_{T^*} \mathbf{r}_1) = g_{T^*}(M_{T^*} \mathbf{r}_2) = g_{T^*}(\mathbf{0})$, and thus

$$g_{T^*}(M_{T^*} \mathbf{r}_1) = g_{T^*}(M_{T^*} \mathbf{r}_2). \quad (5)$$

By condition 1 of the theorem, there exists $\mathbf{u}$ such that $(\alpha'_h)^\top M'_h = (\mathbf{u})^\top M'_{T^*}$. Further, by definition of g , there exists randomness vectors $\mathbf{r}'_1, \mathbf{r}'_2$ for Π' such that $g(M\mathbf{r}_i) = M'\mathbf{r}'_i$ for $i = 1, 2$. Hence,

$$\begin{aligned} (\alpha'_h)^\top g_h(M_h \mathbf{r}_1) &= (\alpha'_h)^\top M'_h \mathbf{r}'_1 = (\mathbf{u})^\top M'_{T^*} \mathbf{r}'_1 = (\mathbf{u})^\top g_{T^*}(M_{T^*} \mathbf{r}_1) \\ &= (\mathbf{u})^\top g_{T^*}(M_{T^*} \mathbf{r}_2) = (\mathbf{u})^\top M'_{T^*} \mathbf{r}'_2 = (\alpha'_h)^\top M'_h \mathbf{r}'_2 = (\alpha'_h)^\top g_h(M_h \mathbf{r}_2). \end{aligned}$$

The first equality in the second line uses Equation (5). We conclude our proof since the above sequence of equations contradicts Equation (4). $\square$

The following corollary from Theorem 11 provides necessary conditions on the convertibility to CNF that are easy to check. In fact, these are conditions only on the access structure, and the share size of a single party.

Corollary 1. *Let Γ denote an access structure on n parties, such that there exists a party h and two minterms T_1, T_2 of size ≥ 2 each, such that $h \in T_1$, $h \in T_2$, $(T_1 \cup T_2) \setminus \{h\}$ is qualified. Let $\Pi = (\mathbb{F}, M \in \mathbb{F}^{m \times k}, \rho)$ be a linear scheme for Γ , and M_h consists of a single row. Then Π is not convertible to $\Pi' = CNF_{\Gamma, \mathbb{F}}$.*

Proof. We apply Theorem 11 to Π, Π' , with $T = T_1 \setminus \{h\}$, $T^* = T_2 \setminus \{h\}$. As to the first condition, we must have

$$\text{Rowspan}(M_h) \cap \text{Rowspan}(M_{T_1 \setminus \{h\}}) = \{\mathbf{0}\},$$

since because M_h is a single row, and it is not spanned by $M_{T_1 \setminus \{h\}}$ since T_1 is a minterm. The same holds for T_2 , so

$$\text{Rowspan}(M_h) \cap \text{Rowspan}(M_{T_2 \setminus \{h\}}) = \{\mathbf{0}\}.$$

Thus,

$$(\alpha_h)^\top M_h \notin \text{Rowspan}(M_h) \cap \text{Rowspan}(M_{T_1 \setminus \{h\}}) + \text{Rowspan}(M_h) \cap \text{Rowspan}(M_{T_2 \setminus \{h\}})$$

for any reconstruction function α for T_1 in Π , and the second condition of Theorem 11 holds.

By CNF construction, every reconstruction function α' for T_1 satisfies $(\alpha')^\top \cdot M_{T_1} \cdot \mathbf{r} = \sum_{H \text{ is a maxterm of } \Gamma} r_H = s$.

To specify our concrete α' , a copy of r_H is contributed by a party $i \in T_1$ that holds it as follows. Each r_H for H for which $T_1 \setminus \{h\} \subseteq H$ are contributed by party h , while every other r_H is contributed by a party in $T_1 \setminus \{h\}$ (picked arbitrarily).

Let us denote the set of maxterms of the first kind by A . Note that A is not empty, since $T_1 \setminus \{h\}$ is unqualified. Also, every r_H for $H \in A$ is indeed held by party h by construction of CNF, as every $H \in A$ does not include h , since T_1 is qualified. Also, each r_H for $H \notin A$ indeed belongs to some $j \in T_1 \setminus \{h\}$, by construction of CNF. In particular, we have $(\alpha'_h)^\top \cdot M'_h \cdot \mathbf{r} = \sum_{H \in A} r_H$.

Since $T_1 \cup T_2 \setminus \{h\}$ is qualified, every r_H for $H \in A$ is held by some party $j \in T_2 \setminus \{h\}$ and therefore $(\alpha'_h)^\top M'_h$ is spanned by the rows of $M'_{T_2 \setminus \{h\}}$. This proves the existence of α' as required by the first condition in Theorem 11. $\square$

Remark 2. From Corollary 1 it follows directly that Shamir secret-sharing scheme is not convertible to CNF, and hence is not maximal. Indeed, any party can be viewed as h , as it obtains a single field element as its share. The condition on the access structure automatically holds for threshold structures.

The following theorem exploits a different property implied by convertability than the previous theorem.

Theorem 12. *Let Γ be an access structure on n parties. Let Π, Π' be linear secret-sharing schemes specified for Γ by MSP $(\mathbb{F}, M \in \mathbb{F}^{m \times k}, \rho)$ and $(\mathbb{F}, M' \in \mathbb{F}^{m' \times k'}, \rho')$, respectively. Let $h \in [n]$ denote a party, and let $\mathcal{T}^h = \{T_1, \dots, T_{v_h}\}$ denote minterms each of which is of size ≥ 2 , and contains h . Let $\{\alpha^{T_i}\}_{i \in [v_h]}$ denote a set of reconstruction functions for sets $T_1, \dots, T_{v_h}$ in Π respectively. Similarly, $\{\alpha'^{T_i}\}_{i \in [v_h]}$ denotes a set of reconstruction functions for $T_1, \dots, T_{v_h}$ in Π' . Assume the following conditions hold:*

1. $A = \{\alpha_h^{T_i}\}_{i \in [v_h]}$ constitutes a set of v_h linearly independent vectors.

2. $A' = \{\alpha'_h{}^{T_i}\}_{i \in [v_h]}$ constitutes a (multi)set of vectors with $\text{rank}(\text{span}(A')) < v_h$.
3. $\text{span}(\{\alpha^\top M_h | \alpha \in A\}) \cap \sum_{i \in [v_h]} (\text{Rowspan}(M_{T_i \setminus h}) \cap \text{Rowspan}(M_h)) = \{\mathbf{0}\}$.

Then Π has no share conversion to Π' .

Proof. Assume for contradiction that a conversion g exists. Fix some party $h \in [n]$ as guaranteed to exist by the theorem. Consider the matrix $\text{Rec}^h = [(\alpha_h^{T_1})^\top M_h; \dots; (\alpha_h^{T_{v_h}})^\top M_h]$. For each $i \in [v_h]$, let $L_i = \text{Rowspan}(M_{T_i \setminus h}) \cap \text{Rowspan}(M_h)$. By condition 1, it is of rank v_h . Therefore, there exists a set $R^h = \{\mathbf{r}^1, \dots, \mathbf{r}^{|\mathbb{F}|^{v_h}}\}$ of randomness values such that

1. $\{\text{Rec}^h \mathbf{r}^j\}_{j \in [|\mathbb{F}|^{v_h}]}$ goes over all distinct vectors in $\mathbb{F}^{v_h}$.
2. For every $j \in [|\mathbb{F}|^{v_h}]$, $i \in [v_h]$, $L_i \mathbf{r}^j = \mathbf{0}$.

Such a set R^h exists, by combining conditions 1 and 3 in the Theorem. Now, let us consider the matrix $\text{Rec}'^h = [(\alpha'_h{}^{T_1})^\top M_h; \dots; (\alpha'_h{}^{T_{v_h}})^\top M_h]$, and let $R'^h = \{\mathbf{r}'^1, \dots, \mathbf{r}'^{|\mathbb{F}|^{v_h}}\}$ denote the set of effective randomness vectors induced by the conversion g (that is $M' \mathbf{r}'^j = g(M \mathbf{r}^j)$). By condition 2 in the Theorem, $\text{rank}(\text{Rec}'^h) < v_h$. Therefore, $\text{span}(\{\text{Rec}'^h \mathbf{r}'^j\}_{j \in [|\mathbb{F}|^{v_h}]})$ is of dimension at most $v_h - 1$, and thus contains at most $|\mathbb{F}|^{v_h-1}$ distinct values. However, by Lemma 1, for every $i \in [v_h]$, $j \in [|\mathbb{F}|^{v_h}]$ we have

$$\text{Rec}'^h[i] \mathbf{r}'^j = \text{Rec}^h[i] \mathbf{r}^j + c^i(\mathbf{0})$$

That is, $c^j(\mathbf{0})$ is some constant, independent of $\mathbf{r}^j$. Therefore,

$$R'^h \mathbf{r}'^j = R^h[j] \mathbf{r}^j + \mathbf{c}$$

where $\mathbf{c}$ is a constant vector. Going over all values of j , on the RHS we obtain a permutation of the set $\mathbb{F}^{v_h}$, but on the LHS, we only get a subset of at most $|\mathbb{F}|^{v_h-1}$ (constituting a linear subspace) – a contradiction. $\square$

The above theorem implies that DNF can only be converted to linear schemes of size at least that of the DNF. More precisely:

Theorem 13. *Let Γ be an access structure over $n > 1$ parties, and $\mathbb{F}$ be a finite field. For simplicity, assume Γ has no redundant parties. Let Π' be a linear secret-sharing scheme specified by MSP $\mathcal{M}' = (\mathbb{F}, M' \in \mathbb{F}^{m' \times k'}, \rho')$ for Γ . If $\mathcal{M} = \text{DNF}_{\Gamma, \mathbb{F}} = (\mathbb{F}, M \in \mathbb{F}^{m \times k}, \rho)$ is convertible to Π' , then ⁹*

$$m' \geq \sum_{T: T \text{ is a minterm of } \Gamma} |T|.$$

⁹ In fact, the number of rows assigned to each party is at least as large as in DNF.

Proof. Consider some $h \in [n]$ that belongs to some minterm T of size at least 2. Let $\mathcal{T}^h = \{T_i \text{ is a minterm} \mid |T_i| \geq 2, h \in T_i\}$, and denote $v_h = |\mathcal{T}^h|$. The (unique, in case of DNF) reconstruction function projection set $A = \{\alpha_h^{T_i}\}$ are all linearly independent by construction of DNF. Namely, each selects a single row in M_h , and $(\alpha_h^{T_i})^\top M_h$ is either of the form $r_{i,h}$ or of the form $s - \sum_{k \in T_i \setminus h} r_{i,k}$, where $r_{i,j}$ is a random element used by the DNF in the additive sub-scheme for minterm T_i and party j .

For a fixed $k \in [v_h]$, by structure of DNF, we have $\text{Rowspan}(M_{T_k \setminus h}) \cap \text{Rowspan}(M_h) = \mathbf{0}$. Therefore, $\sum_{k \in [v_h]} (\text{Rowspan}(M_{T_k \setminus h}) \cap \text{Rowspan}(M_h)) = \{\mathbf{0}\}$. Thus, A satisfies conditions 1 and 3 of Theorem 12 relative to $\mathcal{T}^h$.

It follows from Theorem 12 that A' is of dimension $\geq v_h$. Thus, the matrix $A' M'_h$ (where A' is treated as a matrix constructed from the vectors of A' as rows) also has dimension at least v_h . This implies M'_h itself is of rank at least v_h , which in turn lower bounds its number of rows.

Parties that constitute a singleton minterm, contribute at least 1 to $\text{rank}(M'_h)$, as the party contained in the minterm is not redundant. Going over all parties h , the result follows by changing the order of summation, obtaining.

$$m' \geq \sum_h \sum_{T: T \text{ is a minterm}, h \in T} 1 = \sum_{T: T \text{ is a minterm}} |T| \quad (6)$$

□

Remark 3. Theorem 13, in fact, states that DNF is not convertible to any linear scheme with lower share complexity, in particular, in the Shamir secret sharing scheme. The non-minimality of Shamir follows.

Remark 4. In fact, the above theorem can be fairly easily generalized to Π where $\text{span}(\{\alpha^\top M_h \mid \alpha \in A\}) \cap \sum_{i \in [v_h]} (\text{Rowspan}(M_{T_i \setminus h}) \cap \text{Rowspan}(M_h)) = U$ for some h has a “relatively small” dimension u . Then, we can prove $\dim(\text{span}(A)) - u$ lower bounds $\text{rank}(M'_h)$ if $\Pi \leq \Pi'$. That is, the smaller u relatively to $\dim(\text{span}(A))$ is, the higher the share complexity of party h in Π' must be.

5 A characterization of convertability between linear schemes

Let Γ be an access structure on n parties. Let Π, Π' be linear secret-sharing schemes specified by MSPs $(\mathbb{F}, M \in \mathbb{F}^{m \times k}, \rho)$ and $(\mathbb{F}, M' \in \mathbb{F}^{m' \times k'}, \rho')$, respectively, realizing Γ . We devise a characterization of convertability from Π to Π' by solvability of a certain system of linear equations. Essentially, every solution of the system represents a conversion function g . Namely, for every randomness vector from Π , it defines converted shares for Π' .

The linear system $\mathcal{L}_{\Pi, \Pi'}$. For each randomness vector $\mathbf{r} \in \mathbb{F}^m$ we define a variable $X^{(\mathbf{r})} \in \mathbb{F}^{m'}$, that assumes a value for the purported sharing under M' induced by the share conversion of $M \cdot \mathbf{r}$. The constraints we define are as follows.

- **Locality.** For every $i \in [n]$ and $\mathbf{r}, \mathbf{r}' \in \mathbb{F}^m$ such that $M_i \cdot \mathbf{r} = M_i \cdot \mathbf{r}'$, add the constraint:

$$X_i^{(\mathbf{r})} = X_i^{(\mathbf{r}')}.$$

- **Consistency.** Let $A \subseteq [m']$ be a subset of rows of M' which form a basis of $\text{Rowspan}(M')$. Since $M'[A, \cdot]$ is a basis of $\text{Rowspan}(M')$, there exists a (unique) matrix $H \in \mathbb{F}^{m', |A|}$ such that $H \cdot M'[A, \cdot] = M'$. For every $\mathbf{r} \in \mathbb{F}^m$, we add the constraint

$$H \cdot X^{(\mathbf{r})}[A] = X^{(\mathbf{r})}.$$

- **Correctness.** For each minterm $T \subset [n]$ of Γ do the following: let $\alpha'^T \in \mathbb{F}^{|\rho'^{-1}(T)|}$ be a smallest reconstruction vector for T under some arbitrary ordering of $\mathbb{F}^{|\rho'^{-1}(T)|}$. For every $s \in \mathbb{F}$, and every $\mathbf{r}$ such that $\mathbf{r}[1] = s$, add the constraint

$$\left(\alpha'^T\right)^T \cdot X_T^{(\mathbf{r})} = s.$$

Remark 5. The characterization may be easily extended to non-linear Π with $S = \mathbb{F}$, keeping the system linear.

Theorem 14. *Let Γ be an access structure on n parties. Let Π, Π' be linear secret-sharing schemes specified by MSPs $(\mathbb{F}, M \in \mathbb{F}^{m \times k}, \rho)$ and $(\mathbb{F}, M' \in \mathbb{F}^{m' \times k'}, \rho')$, respectively, realizing Γ . Then, Π is convertible to Π' if and only if the linear system $\mathcal{L}_{\Pi, \Pi'}$ is solvable.*

Proof. In one direction, assume a share conversion function g converting Π to Π' exists. Then, let $X^{(\mathbf{r})} = g_i(M_i \cdot \mathbf{r})$ for each $\mathbf{r} \in \mathbb{F}^m$. The locality constraint is satisfied by locality of g_i (the fact that sh_i' depends only on sh_i). Correctness follows from the correctness of share conversion. Consistency is satisfied since, for each $\mathbf{r} \in \mathbb{F}^m$, there exists $\mathbf{r}' \in \mathbb{F}^{m'}$ such that $g(M \cdot \mathbf{r}) = M' \cdot \mathbf{r}'$. The consistency now follows from the property of M' .

In the other direction, assume a solution to the system exists. Then, $g = (g_1, \dots, g_n)$ where for every $i \in [n]$, and every value $\text{sh}_i \in \mathbb{F}^{|\rho^{-1}(i)|}$, choose some $\mathbf{r} \in \mathbb{F}^m$ such that $M_i \cdot \mathbf{r} = \text{sh}_i$, and set $g_i(\text{sh}_i) = X_i^{(\mathbf{r})}$. By locality, g is well-defined. By consistency, for every $\text{sh} = M \cdot \mathbf{r}$ such that $\mathbf{r}[1] = s$, $\text{sh}' = g(\text{sh}) = M' \cdot \mathbf{r}'$ for a unique $\mathbf{r}' \in \mathbb{F}^{m'}$. By Correctness, $\mathbf{r}'[1] = s$. $\square$

Theorems 11 and 12 may be viewed as a result of an inconsistency in the characterization equations. In Theorem 11, the inconsistent subset of equations supported only on $X^{(\mathbf{r}^1)}, X^{(\mathbf{r}^2)}$ for a pair $\mathbf{r}_1, \mathbf{r}_2$ as chosen in the proof. In Theorem 11, the inconsistency involves a potentially large set of $\mathbf{r}^j$'s, already. Lemma 1 relies on equations of all three types to derive its conclusion.

Future work. We believe that there exist additional types of "inconsistencies" in the linear equations in the characterization that may result in proving non-existence of a conversion from between pairs of linear schemes Π to Π' , and it is an interesting open question to list all possible types of inconsistencies, and thereby make the above characterization easier to apply. Most importantly, we

wish to understand when conversions between linear schemes *do* exist. Another interesting open question on convertability between pairs of linear schemes is understanding whether it is helpful to have non-linear conversion functions g .

6 Impossibility of conversion to CNF for general schemes

In this section, we introduce the class of non-degenerate secret-sharing schemes, which includes CNF and Shamir schemes, and, using properties of non-degenerate schemes, we prove a necessary condition of convertability to CNF from any (not necessarily linear) secret-sharing scheme.

Definition 18. Let Π be a secret-sharing scheme with secret domain S and randomness domain R realizing an access structure Γ . Π is non-degenerate if the following condition is met: If Π' is a secret-sharing scheme with secret domain S and randomness domain R' realizing Γ such that, for all $s \in S, r' \in R'$, there exists $r \in R$ such that $\Pi'(s; r') = \Pi(s; r)$, then

$$(\Pi'(s; r') | r' \leftarrow R') \equiv (\Pi(s; r) | r \leftarrow R), \forall s \in S. \quad (7)$$

Suppose Π is a non-degenerate secret-sharing scheme. If a secret-sharing scheme Π' is locally convertible to Π , then the secret-sharing scheme induced by the applying the share conversion function to Π' coincides with Π .

Proposition 2. Let Π be a non-degenerate secret-sharing scheme with secret domain S and randomness domain R realizing access structure Γ over n parties. Suppose Π' be a secret-sharing scheme with same secret domain and access structure and randomness domain R' that admits share conversion to Π using a share conversion function $g = (g_1, \dots, g_n)$. Then, for all $s \in S$,

$$(g(\Pi'(s; r')) | r' \leftarrow R') \equiv (\Pi(s; r) | r \leftarrow R). \quad (8)$$

Proof. Consider the secret-sharing scheme in which $s \in S$ is secret shared as $(sh_1, \dots, sh_n) = g(\Pi'(s; r'))$ where $r' \leftarrow R'$. Since g is a share conversion function that converts Π' to Π , this induces secret-sharing scheme with secret domain S realizing the access structure Γ . Further, for each $s \in S$ and $r' \in R'$, $g(\Pi'(s; r')) = \Pi(s; r)$ for some $r \in R$. The proposition now follows from the fact that Π is a non-degenerate secret-sharing scheme (See definition 18). $\square$

Theorem 15. For any finite group $\mathbb{G}$, and access structure Γ over n parties, the CNF secret-sharing scheme for secrets in $\mathbb{G}$ realizing Γ is non-degenerate.

Proof. Denote the set of max-terms of Γ is by $\mathcal{F} = \{F \subseteq [n] : F \notin \Gamma, F' \supset F \implies F' \in \Gamma\}$. Let Π be any secret-sharing scheme with secret domain $\mathbb{G}$ and randomness domain R realizing the access structure Γ that satisfies the following condition: for any $s \in \mathbb{G}$ and $r \in R$, let $(sh_1, \dots, sh_n) = \Pi(s; r)$. Then, there exist $\{\gamma_F \in \mathbb{G} : F \in \mathcal{F}\}$ such that $\sum_F \gamma_F = s$ and $sh_i = \{\gamma_F : F \in \mathcal{F}, i \notin F\}$ for all $i \in [n]$. We will show that, for any s , when $r \leftarrow R$, $\Pi(s; r)$ is identically

distributed as secret-sharing of s under the CNF secret-sharing scheme. This will prove the theorem.

Let $s \leftarrow \mathbb{G}$, $r \leftarrow R$ and $(\text{sh}_1, \dots, \text{sh}_n) = \Pi(s; r)$. Let $\text{sh}_i = \{\gamma_F : i \notin F \in \mathcal{F}_{\max}\}$ for each $i \in [n]$. We claim, for each $F \in \mathcal{F}_{\max}$,

$$H(\gamma_F \mid \{\gamma_{F'}\}_{F' \neq F}) = \log |\mathbb{G}|. \quad (9)$$

We first prove the theorem assuming the above equality: Let $\{F : F \in \mathcal{F}, F \neq F^*\} = \{F_1, \dots, F_k\}$, where F^* is some arbitrary member of $\mathcal{F}$. Then, by chain rule of entropy,

$$\begin{aligned} H(\{\gamma_F\}_{F \neq F^*}) &= H(\gamma_{F_1}, \dots, \gamma_{F_k}) \\ &= \sum_{i=1}^k H(\gamma_{F_i} \mid \gamma_{F_1}, \dots, \gamma_{F_{i-1}}) \geq \sum_{i=1}^k H(\gamma_{F_i} \mid \{\gamma_F\}_{F \neq F_i}) = k \log |\mathbb{G}|. \end{aligned}$$

But, each γ_F is distributed over $\mathbb{G}$. Hence, $H(\{\gamma_F\}_{F \neq F^*}) = k \log |\mathbb{G}|$ if and only if each γ_F , where $F \neq F^*$, is uniformly and independently distributed over $\mathbb{G}$. Finally, by the correctness of Π , $\gamma_{F^*} = s - \sum_{F \neq F^*} \gamma_F$. Thus, as required in the statement of the theorem, for each $s \in \mathbb{G}$,

$$\begin{aligned} (\text{sh}_1, \dots, \text{sh}_n \mid r \leftarrow R, (\text{sh}_1, \dots, \text{sh}_n) = \Pi(s; r)) &\equiv \\ \left(\text{sh}_1, \dots, \text{sh}_n \mid \begin{array}{l} \gamma_F \leftarrow \mathbb{G}, \forall F \in \mathcal{F} \text{ subj to } \sum_{F \in \mathcal{F}} \gamma_F = s \\ \text{sh}_i = \{\gamma_F : i \in F, F \in \mathcal{F}\} \end{array} \right). \end{aligned}$$

We conclude by showing eq. (9). Fix $F \in \mathcal{F}$. By definition of maximal forbidden sets, for each $F' \neq F$, there exists some $i \in F$ such that $i \notin F'$; in other words, $\gamma_{F'} \in \text{sh}_i$ for some $i \in F$. But then,

$$H(s \mid \{\gamma_{F'}\}_{F' \neq F}) = H(s \mid \{\text{sh}_i\}_{i \in F}) = H(s) = \log |\mathbb{G}|,$$

where the second equality follows from F being a forbidden set, and the perfect privacy of Π ; third equality follows from s be uniformly distributed over $\mathbb{G}$. Furthermore, $H(s \mid \{\gamma_{F'}\}_{F' \in \mathcal{F}}) = 0$ since $\sum_{F' \in \mathcal{F}} \gamma_{F'} = s$. Hence,

$$\begin{aligned} H(\gamma_F \mid \{\gamma_{F'}\}_{F' \neq F}) &= H(\gamma_F \mid \{\gamma_{F'}\}_{F' \neq F}) + H(s \mid \{\gamma_{F'}\}_{F' \neq F}, \gamma_F) \\ &= H(s, \gamma_F \mid \{\gamma_{F'}\}_{F' \neq F}) \geq H(s \mid \{\gamma_{F'}\}_{F' \neq F}) = \log |\mathbb{G}|. \end{aligned}$$

This proves eq. (9) concluding the proof. $\square$

Theorem 16. *For any finite field $\mathbb{F}$ such that $|\mathbb{F}| > n$, and $1 \leq t \leq n$, a t -private n -party Shamir secret-sharing scheme over $\mathbb{F}$ is non-degenerate.*

Proof. Let Π be any t -private n -party threshold secret-sharing scheme with secret domain $\mathbb{F}$ and randomness domain R that satisfies the following condition: for any $s \in \mathbb{F}$ and $r \in R$, let $(\text{sh}_1, \dots, \text{sh}_n) = \Pi(s; r)$. Then, there exists a polynomial $p(x)$ of degree at most t such that $p(0) = s$ and $\text{sh}_i = p(i)$ for each $i \in [n]$. We will show that, for any s , when $r \leftarrow R$, $\Pi(s; r)$ is identically distributed as

secret-sharing of s under the t -private n -party Shamir secret-sharing scheme over $\mathbb{F}$. This will prove the theorem.

Let $s \leftarrow \mathbb{G}$, $r \leftarrow R$ and $(\text{sh}_1, \dots, \text{sh}_n) = \Pi(s; r)$. We claim, for each $i \in [t+1]$,

$$H(\text{sh}_i \mid \text{sh}_1, \dots, \text{sh}_{i-1}, \text{sh}_{i+1}, \dots, \text{sh}_{t+1}) = \log |\mathbb{F}|. \quad (10)$$

We first prove the theorem assuming the above equality. By chain rule of entropy,

$$\begin{aligned} H(\text{sh}_1, \dots, \text{sh}_t) &= \sum_{i=1}^t H(\text{sh}_i \mid \text{sh}_1, \dots, \text{sh}_{i-1}) \geq \\ &\geq \sum_{i=1}^t H(\text{sh}_i \mid \text{sh}_1, \dots, \text{sh}_{i-1}, \text{sh}_{i+1}, \dots, \text{sh}_{t+1}) = t \log |\mathbb{F}|. \end{aligned}$$

Since each sh_i is distributed over $\mathbb{F}$, this implies that $\{\text{sh}_i\}_{1 \leq i \leq t}$ is uniformly and independently distributed over $\mathbb{F}$. Finally, by the correctness of Π , there exists a polynomial $p(x)$ of degree at most t -i.e., $p(s) \leftarrow \mathbb{F}^{\leq t}[x]$ such that $\text{sh}_i = p(i)$ for each $1 \leq i \leq n$ and $p(0) = s$. Since this polynomial is uniquely determined by s and $\{\text{sh}_i\}_{1 \leq i \leq t}$, for each $s \in \mathbb{G}$, as required by the theorem,

$$\begin{aligned} (\text{sh}_1, \dots, \text{sh}_n \mid r \leftarrow R, (\text{sh}_1, \dots, \text{sh}_n) = \Pi(s; r)) &\equiv \\ (\text{sh}_1, \dots, \text{sh}_n \mid p(x) \leftarrow \mathbb{F}^{\leq d}[x] \text{ subj to } p(0) = s, \text{sh}_i = p(i)) \end{aligned}$$

We conclude by showing eq. (10). Fix $1 \leq i \leq t+1$.

$$H(s \mid \text{sh}_1, \dots, \text{sh}_{i-1}, \text{sh}_{i+1}, \dots, \text{sh}_{t+1}) = H(s) = \log |\mathbb{F}|,$$

where the first equality follows from the perfect privacy of Π ; second equality follows from s be uniformly distributed over $\mathbb{F}$. Furthermore, $\text{sh}_1, \dots, \text{sh}_{t+1}$ uniquely determine the polynomial $p(x)$ of degree at most d such that $p(0) = s$ and $\text{sh}_i = p(i)$ for all $i \in [n]$. Hence, $H(s \mid \text{sh}_1, \dots, \text{sh}_{t+1}) = 0$. Hence,

$$\begin{aligned} &H(\text{sh}_i \mid \text{sh}_1, \dots, \text{sh}_{i-1}, \text{sh}_{i+1}, \dots, \text{sh}_{t+1}) \\ &= H(\text{sh}_i \mid \text{sh}_1, \dots, \text{sh}_{i-1}, \text{sh}_{i+1}, \dots, \text{sh}_{t+1}) + H(s \mid \text{sh}_1, \dots, \text{sh}_{t+1}) \\ &= H(s, \text{sh}_i \mid \text{sh}_1, \dots, \text{sh}_{i-1}, \text{sh}_{i+1}, \dots, \text{sh}_{t+1}) \\ &\geq H(s \mid \text{sh}_1, \dots, \text{sh}_{i-1}, \text{sh}_{i+1}, \dots, \text{sh}_{t+1}) = \log |\mathbb{F}|. \end{aligned}$$

This proves eq. (10) concluding the proof. $\square$

Theorem 17. *Let Π be a secret-sharing scheme with secret domain $\mathbb{G}$ and randomness domain R realizing an n -party access structure Γ . There is a share conversion from Π to CNF secret-sharing over $\mathbb{G}$ realizing Γ only if, for each $i \in [n]$, size of the share i in Π is at least $\log |\mathbb{G}| \cdot |\{F \in \mathcal{F} \text{ s.t. } i \notin F\}|$, where $\mathcal{F}$ is the set of all maximal forbidden sets associated with Γ .*

Proof. By theorem 15, the CNF secret-sharing scheme is non-degenerate. Let $g = (g_1, \dots, g_n)$ be the share conversion function that induces the share conversion

from Π to the CNF secret-sharing scheme. By proposition 2, for any $s \in \mathbb{G}$, when $r \leftarrow R$, $g(\Pi(s; r))$ is identically distributed as CNF secret-sharing of s . Hence, $g_i(\Pi(s; r))$ corresponds to the share of party i in CNF secret-sharing: $\{\gamma_F : F \in \mathcal{F}, i \notin F\}$ where γ_F is uniformly chosen from $\mathbb{G}$ for each $F \in \mathcal{F}$ subject to $\sum_F \gamma_F = s$. Theorem follows immediately from this observation. $\square$

7 Results for Evolving Linear Secret-Sharing Schemes

In this section, we extend the notion of Monotone Span Programs and the induced notion of a linear secret-sharing scheme to the evolving setting. We then apply our impossibility results obtained in Sections 4 and 6 for the finite case to study the convertibility hierarchy in this setting.

Monotone span programs [15] were used to construct linear secret-sharing schemes in [4]. In this section, we extend Definition 8 to define infinite monotone span programs and cast a few constructions from the literature as instances of this notion. We define the product of an infinite matrix $K \in \mathbb{F}^{[n] \times \mathbb{N}^+}$ by a finite vector $\mathbf{r} \in \mathbb{F}^{[m]}$ as $K'\mathbf{r}$, where K' is obtained by keeping the first m columns of K . We will typically use such products for matrices where all but the first m columns are 0. We next introduce extension of MSP to the infinite case.¹⁰

Definition 19 (Infinite Monotone Span Program– IMSP). *An IMSP is a triple $\mathcal{M} = (\mathbb{F}, M, \rho)$, where $\mathbb{F}$ is a finite field, $M \in \mathbb{F}^{\mathbb{N} \times \mathbb{N}}$ is an infinite matrix over $\mathbb{F}$, and $\rho : \mathbb{N}^+ \rightarrow \mathbb{N}^+$ labels each row of M by a party. There is finite number of non zero elements each row in M , and $\rho^{-1}(x)$ is finite for every $x \in \mathbb{N}^+$, that is, each party get finite number of rows (shares). For any finite set $A \subseteq [n]$ of party indices, let M_A denote the sub-matrix obtained by restricting M to the rows i , with $\rho(i) \in A$. We say that $\mathcal{M}$ accepts B if the rows of M_B span the vector $\mathbf{e}_1 = (1, 0, 0, \dots)$.¹¹ We say that $\mathcal{M}$ implements an evolving access structure Γ if $\mathcal{M}$ accepts a set B if and only if $B \in \Gamma$.*

In the following theorem, we generalize the MSP-based linear secret-sharing schemes to the evolving setting, essentially giving each party the linear combinations of a randomness vector (that also defines the secret s), as specified by the IMSP. As in the finite case, every finite subset $A \subseteq \mathbb{N}^+$ either reconstructs the secret, or learns nothing about it.

¹⁰ We use a ‘working definition’ of linear evolving secret-sharing schemes specified by an IMSP, which is a natural extension of the finite case. An arguably more intuitive definition is requiring that all shares are linear combinations of the r_i ’s and s (over a field $\mathbb{F}$) without the restriction on reconstruction. Beimel has demonstrated in [3] that linear schemes imply MSPs of similar share complexity, so the definitions are equivalent. We do not demonstrate such a result in this paper, as it is outside of its main scope, but an analog of this result would be useful to demonstrate in a future paper focusing on the theory of linear evolving schemes.

¹¹ Generally, it is possible to take any vector with a finite positive number of non-zero entries as a target vector.

Theorem 18. *Let $\mathcal{M} = (\mathbb{F}, M, \rho)$ be an IMSP accepting an access structure Γ . Then, Construction 19 instantiated with $\mathcal{M}$ implements Γ .*

Proof. Given an IMSP $\mathcal{M} = (\mathbb{F}, M, \rho)$, we define an *evolving linear secret-sharing scheme* as follows:

For a finite set of parties A , denote by $C_A = \{j | \exists i, \rho^{-1}(i) \in A, M_{i,j} \neq 0\}$, the set of non-zero entries it holds.

Construction 19. *Consider an IMSP $\mathcal{M}(\mathbb{F}, M, \rho)$.*

- *INPUT:* a secret $s \in \mathbb{F}$.
We determine $r_0 = s$ and define $\mathbf{r} = (s, r_1, r_2 \dots)$.
- *SHARE:* To generate sh_i the dealer does the following:
 1. Gets as input $(s, \text{sh}_1, \dots, \text{sh}_{i-1})$, that is, a secret s and the shares of parties $p_1, \dots, p_{i-1}$. For convenience, we assume it in fact receives the set of $r_1, \dots, r_j$'s sampled so far, for $j = \max(C_{[i-1]})$. It samples random independent elements $r_{j+1}, \dots, r_{j+d} \in \mathbb{F}$ for $j + d = \max(C_{[i]})$.
 2. Set $\text{sh}_i = M_i \mathbf{r}$, where $\mathbf{r}$ is the prefix of $\mathbf{r}$ sampled so far.
- *RECON:* Let α denote the (finite) reconstruction vector such that $\alpha^T M_B = \mathbf{e}_1$. Return $\langle \alpha, \Pi(B) \rangle$.

Correctness and privacy of this scheme follow similarly to the correctness and privacy of the standard scheme in [4], by considering the finite submatrices corresponding to finite subsets $[n]$ of parties. □

The proof of this theorem re-interprets the construction of [1] as an IMSP based scheme. We define *evolving linear secret-sharing schemes* as the set of schemes so specified by an IMSP. In Appendix A we describe constructions for the evolving threshold, and for the evolving undirected *st*-connectivity access structures families as instances of IMSP.

Theorem 20. *Let Γ denote an evolving access structure. Then GIDT [1, Construction 3.9] for $S = \mathbb{F}_2$ and Γ instantiated so that edge predicates are implemented by linear schemes over $\mathbb{F}_2$ is (evolving) linear.*

The proof immediately follows from observing the GIDT [1] given in Definition 14 as Construction 10.

The following theorem states that for a large class of evolving access structures there is no minimal linear evolving secret-sharing scheme. This proof follows from Theorem 11 applied to any specific evolving scheme and a tailor crafted GIDT scheme [1, Construction 3.9].

Theorem 21. *Consider an evolving access structure Γ such that there exists $\tilde{h} \in [n]$, and an infinite collection of minterms $A = \{T_i\}_{i \in \mathbb{N}}$ where $\tilde{h} \in T_i$ for all $i \in \mathbb{N}$. Then, for any linear scheme $\tilde{\Pi}$ specified by M over $\mathbb{F}_2$, there exists an evolving linear scheme $\tilde{\Pi}'$ specified by M' over $\mathbb{F}_2$ for Γ , such that $\tilde{\Pi}'$ is not convertible to $\tilde{\Pi}$.*

Proof. First observe that, by the assumption in Theorem 11, and the pigeon hole principle, there exists a pair of minterms T_i, T_j of size at least 2 each with $\tilde{h} \in T_i \cap T_j$, and reconstruction vectors α^i, α^j corresponding to T_i, T_j respectively, that satisfy $\alpha_{\tilde{h}}^i = \alpha_{\tilde{h}}^j$. Let n is the last party in $T_i \cup T_j$, and let $d \in T_i$ denote a party in $T_i \setminus (T_j \cup \{\tilde{h}\})$. First, observe that $(\alpha_d^i)^\top \cdot M_d$ is not spanned by $T_i \setminus \{d\}$ (or else $T_i \setminus \{d\}$ would be qualified). As $(\alpha^i)^\top \cdot M_{T_i} = (\alpha^i)^\top \cdot M_{T_j} = \mathbf{e}_i$, we have

$$(\alpha_d^i)^\top \cdot M_d + (\alpha_{T_i \setminus \{d, \tilde{h}\}}^i)^\top \cdot M_{T_i \setminus \{d, \tilde{h}\}} + (\alpha_{\tilde{h}}^i)^\top \cdot M_{\tilde{h}} = (\alpha_{T_j \setminus \{\tilde{h}\}}^i)^\top \cdot M_{T_j \setminus \{\tilde{h}\}} + (\alpha_{\tilde{h}}^j)^\top \cdot M_{\tilde{h}} \quad (11)$$

Here we use the fact that $\alpha_{\tilde{h}}^i = \alpha_{\tilde{h}}^j$. Thus, it follows from Equation (11) that

$$(\alpha_d^i)^\top \cdot M_d = (\alpha_{T_j \setminus \{\tilde{h}\}}^j)^\top \cdot M_{T_j \setminus \{\tilde{h}\} \setminus F} - (\alpha_{T_i \setminus \{d, \tilde{h}\}}^i)^\top \cdot M_{T_i \setminus \{d, \tilde{h}\}}. \quad (12)$$

This implies that $(\alpha_d^i)^\top \cdot M_d$ is spanned by the rows of $M_{(T_i \cup T_j) \setminus \{d, \tilde{h}\}}$. Now, let us take $\tilde{\Pi}'$ the GIDT-based scheme from Construction 10 where the first generation is $[n]$, and the predicate at the edge going from the root into a leaf is implemented via a $DNF_{\Gamma, \mathbb{F}_2}$ scheme (other edges' predicates may be implemented by an arbitrary linear scheme over $\mathbb{F}_2$). Let $\Pi' = \tilde{\Pi}$, $\Pi = \tilde{\Pi}'$, $T = T_i \setminus \{d\}$, $T' = (T_j \cup T_i) \setminus d, \tilde{h}$, $h = d$, $\alpha' = \alpha^i$. Let α be a reconstruction function that picks from M_{T_i} the rows corresponding to $\mathbf{r}_{T_i, w}$'s (each party $w \in T_i$ holds a single matrix row of the form $\mathbf{r}_{T_i, w}$, which is either a fresh random value or $s - \sum_{w > w'} \mathbf{r}_{T_i, w'}$, for the minimal $w' \in T_i$). The conditions of Thm. 11 are satisfied, implying that $\tilde{\Pi}'$ is not convertible to $\tilde{\Pi}$. This follows directly from the above equations, and the following observation on DNF and the evolving scheme.

Note that $(\alpha_d)^\top \cdot M'_d \notin \text{Rowspan}(M'_d) \cap \text{Rowspan}(M'_{T_i \setminus \{d\}}) + \text{Rowspan}(M'_d) \cap \text{Rowspan}((M'_{T_i \cup T_j}) - d)$. To see this, observe that the DNF part contributes to $\text{Rowspan}(M'_d) \cap \text{Rowspan}(M'_{T_i \setminus \{d\}})$ only the vector $\mathbf{0}$. The non-DNF parts (corresponding to edges going into non-leaf nodes), contribute a vector space V consisting entirely of vectors spanned by a set of random elements (vectors) disjoint of the randomness used by the DNF part (including s). This is the case, as the other edges share fresh random labels (of the corresponding edges), using fresh randomness. $\text{Rowspan}(M'_d) \cap \text{Rowspan}(M'_{(T_i \cup T_j) \setminus \{d, \tilde{h}\}})$ does not span $(\alpha_d)^\top \cdot M'_d$ by properties of DNF, and the fact that $(T_i \cup T_j) \setminus \{d, \tilde{h}\}$ misses both d and $\tilde{h}$ from T_i (even if it contains other minterms, thus spanning s). Overall, $V + \text{Rowspan}(M_{(T_i \cup T_j) \setminus \{d, \tilde{h}\}})$ therefore does not contain $(\alpha_d)^\top \cdot M'_d$. This concludes the proof. $\square$

Next, using results obtained in Section 6, we prove the absence of a maximal evolving scheme in a wide class of evolving secret-sharing schemes, even not necessarily linear.

Definition 20 (Trivial Evolving Access Structures). *An evolving access structure Γ is said to be trivial if there exists $N \in \mathbb{N}$ such that, for all $n > N$, $\{n\} \in \Gamma$ or for all finite set A , $A \in \Gamma$ only if $A \setminus \{n\} \in \Gamma$.*

Theorem 22. *Any non-trivial evolving access structure Γ and finite field $\mathbb{F}$ has no maximal evolving secret-sharing scheme over $\mathbb{F}$.*

Proof. We will use the following claim to prove the theorem.

Claim 4. If Γ is non-trivial, then for any $k \in \mathbb{N}$, there exists $n \in \mathbb{N}$ such that, when $\mathcal{F}_n$ is the set of max-terms of Γ_n , $|\{F \in \mathcal{F}_n : 1 \notin F\}| \geq k$.

Before we prove the claim, we will use this claim to prove the theorem. Let Π be a purported maximal secret-sharing scheme for one bit secrets realizing the access structure Γ . Let $|\text{sh}_1|$ be the share size of the share assigned to party 1 by Π . By the above claim, there exists n such that, when $\mathcal{F}_n$ is the set of max-terms of Γ_n , $|\{F \in \mathcal{F}_n : 1 \notin F\}| > |\text{sh}_1|$. Consider the GIDT-based construction Π' for Γ of [1] with the first generation consisting of n parties, and a CNF implementation of Γ_n , in the edge going from the root to a leaf. By Theorem 17, Π does not have a share conversion to Π' since the size of share i is less than $|\{F \in \mathcal{F}_n : 1 \notin F\}| > |\text{sh}_1|$. We conclude the proof by proving the claim.

Proof of the claim. Assume towards a contradiction that there exist k, n^* such that, for all $n \geq n^*$, $|\{F \in \mathcal{F}_n : 1 \notin F\}| = k$, where $\mathcal{F}_n$ are maxterms of Γ_n .

We will first show that there exists $\tilde{n} > n^*$ such that $\{\tilde{n}\} \notin \Gamma_{\tilde{n}}$, and there exists a max-term $\tilde{F} \in \mathcal{F}_{\tilde{n}-1}$ such that $\tilde{F} \cup \{\tilde{n}\} \in \Gamma_{\tilde{n}}$. Since Γ is non-trivial, there exists $n > n^*$ such that $\{n\} \notin \Gamma$, and for some finite set $A \neq \{n\}$, $A \setminus \{n\} \notin \Gamma$ but $A \in \Gamma$. Let $\tilde{n} > n^*$ be the largest value in A . Note, $\{\tilde{n}\} \notin \Gamma$. This can be seen as follows: the statement holds if $\tilde{n} = n$, otherwise $\tilde{n} \in A \setminus \{n\} \notin \Gamma$. Since $\{\tilde{n}\}$ is not authorized, $\max(A) = \tilde{n}$, and $A \in \Gamma$, we reach the following conclusions: (i). $\{\tilde{n}\} \notin \Gamma_{\tilde{n}}$, (ii). there exists $\tilde{F} \supseteq A \setminus \{\tilde{n}\}$ such that $\tilde{F} \in \mathcal{F}_{\tilde{n}-1}$ and $\tilde{F} \cup \{\tilde{n}\} \in \Gamma_{\tilde{n}}$.

Let $S_{\tilde{n}-1} = \{F \in \mathcal{F}_{\tilde{n}-1} : 1 \notin F\}$, and $S_{\tilde{n}} = \{F \in \mathcal{F}_{\tilde{n}} : 1 \notin F\}$. We will show that $|S_{\tilde{n}}| > |S_{\tilde{n}-1}|$, reaching a contradiction. For this, consider the map τ that takes any $F \in S_{\tilde{n}-1}$ to $\tau(F) = F \cup \{\tilde{n}\}$ if $F \cup \{\tilde{n}\} \in \mathcal{F}_{\tilde{n}}$ and to $\tau(F) = F$ otherwise. Observe that, for all $F \in S_{\tilde{n}-1}$, $\tau(F) \in \mathcal{F}_{\tilde{n}}$, and τ is one-to-one. Suppose $1 \notin \tilde{F}$. Then, since $1 \notin \tilde{F} \cup \{\tilde{n}\} \in \Gamma_{\tilde{n}}$, there exists a non-empty F' such that $\{\tilde{n}\} \in F' \subset \tilde{F} \cup \{\tilde{n}\}$ and $F' \in S_{\tilde{n}}$. Since $\tau(\tilde{F}) = \tilde{F}$, and $F' \setminus \{\tilde{n}\} \subset \tilde{F}$, F' is not in the co-domain of τ . Since τ is one-to-one, we conclude $|S_{\tilde{n}}| > |S_{\tilde{n}-1}|$.

Next, let $1 \in \tilde{F}$. Since $\tilde{F} \cup \{\tilde{n}\} \in \Gamma_{\tilde{n}}$, there exists F' such that $1 \notin F' \subset \tilde{F} \cup \{\tilde{n}\}$ such that $F' \in \mathcal{F}_{\tilde{n}}$. Since $1 \notin F'$, $F' \in S_{\tilde{n}}$. However, we observe that for any $F \in S_{\tilde{n}-1}$, and $F' \setminus \{\tilde{n}\} \subsetneq \tilde{F} \setminus \{1\}$, holds $\tilde{F} \setminus \{1\} \not\supseteq F$. Hence, there is no $F \in S_{\tilde{n}-1}$ such that $F' \setminus \{\tilde{n}\} = F$. In other words, $F' \setminus \{\tilde{n}\} \notin S_{\tilde{n}-1}$. Thus, $F' \in S_{\tilde{n}}$ but F' is not in the co-domain of τ . Since τ is one-to-one, we conclude $|S_{\tilde{n}}| > |S_{\tilde{n}-1}|$. $\square$

This completes the proof of the theorem. $\square$

8 Extensions and applications

Above, we considered secret-share conversion for schemes defined over the same field. In this section, we discover the possibility to perform share conversion

between schemes over different fields. We show that the most of our impossibility results is applicable to the case when the source and target schemes are defined over the fields of the same characteristic, and when characteristics are different, the conversion is not possible for many access structures.

8.1 Extending impossibility results to schemes over different fields of the same characteristics

In this section, we extend our impossibility results following from Theorem 11 to secret sharing schemes defined over distinct fields of the same characteristic. For this, we slightly restate Lemma 1 and Theorem 11 such that Π and Π' are two secret sharing schemes defined over $\mathbb{F}$ and $\mathbb{F}'$, respectively, which are the extension fields of $\mathbb{F}_p$.

Observation 23. *Lemma 1 holds for secret-sharing schemes Π defined by MSP $(\mathbb{F}, M \in \mathbb{F}^{m \times k}, \rho)$ and Π' defined by MSP $(\mathbb{F}', M' \in \mathbb{F}'^{m' \times k'}, \rho')$ with the secret domain $\mathbb{F}_p$, where $\mathbb{F}$ and $\mathbb{F}'$ are extension fields of $\mathbb{F}_p$ for prime p .*

Proof. The proof is the same as the proof of Lemma 1, with the difference that (1) holds over $\mathbb{F}'$, Equation (2) is over $\mathbb{F}^*$ which is the extension field for both $\mathbb{F}$ and $\mathbb{F}'$, and Equation (3) is over $\mathbb{F}'$ (where function c is over $\mathbb{F}^*$ with the output in $\mathbb{F}'$). $\square$

Observation 24. *Theorem 11 holds for secret-sharing schemes Π defined by MSP $(\mathbb{F}, M \in \mathbb{F}^{m \times k}, \rho)$ and Π' defined by MSP $(\mathbb{F}', M' \in \mathbb{F}'^{m' \times k'}, \rho')$ with the secret domain $\mathbb{F}_p$, where $\mathbb{F}$ and $\mathbb{F}'$ are extension fields of $\mathbb{F}_p$ for prime p .*

Proof. The proof of this observation is the same as the proof of Theorem 11, where all the equations before (4) are over $\mathbb{F}$. To obtain Equation (4), we apply Theorem 11 under Observation 23 which facilitates the transition to the field $\mathbb{F}'$. Equations (4), (5), and all the following proof are constructed over $\mathbb{F}'$. $\square$

We rely on the above observation to extend our impossibility result in Corollary 1 for conversion from certain linear schemes to CNF. In the observation below, both source and target schemes work for sharing secrets in the larger domains $\mathbb{F}$, $\mathbb{F}'$, but are used only to share secrets in the base field $\mathbb{F}_p$. In fact, any such scheme may be viewed as a scheme over the smaller field S , where each party, including h receives several field elements, which correspond to operations over $\mathbb{F}_p$. Shamir over a large extension field of $\mathbb{F}_p$ is one example of such a scheme.

Observation 25. *Corollary 1 holds for secret-sharing schemes Π defined by MSP $(\mathbb{F}, M \in \mathbb{F}^{m \times k}, \rho)$ and $\Pi' = \text{CNF}_{\Gamma, \mathbb{F}'}$ with the secret domain $\mathbb{F}_p$, where $\mathbb{F}$ and $\mathbb{F}'$ are extension fields of $\mathbb{F}_p$ for prime p .*

Proof. The proof of this observation is the same as the proof of Corollary 1, where Theorem 11 is applied under Observation 23. $\square$

Note that the above result is not subsumed by Theorem 17, as it allows $\mathbb{F}$ to be much larger than $\mathbb{F}'$.

8.2 Negative result for the inter-field conversion.

It is often useful in applications to perform share conversion between schemes over different fields of different characteristic. A natural choice of a secret domain for such a conversion is $\{0, 1\}$, as these values belong to all finite fields. Furthermore, these values can be viewed as bits, which is the most useful setting for most MPC protocols. In this section, we show that a local conversion, in general, is not possible (even from CNF) for many access structures. However, below we show a specially tailored (leaky) secret-sharing scheme over field $\mathbb{Z}_p$ which allows local conversion to a different field $\mathbb{Z}_q$ for $q < n/2$ for (n, n) -threshold.

Next, we observe that for all pairs $p \neq q$, and many access structures Γ , one can not convert from $CNF_{\Gamma, \mathbb{F}_p}$ to $\Pi_{\Gamma, \mathbb{F}_q}$, where $\Pi_{\Gamma, \mathbb{F}_q}$ is any linear scheme over q for that share. More precisely, we have

Theorem 26. *Let Γ denote an access structure for $n > 1$ parties, such that for all maxterms B_1, B_2 , $B_1 \cup B_2 = [n]$.¹² Let $p \neq q$ be primes, and $CNF_{\Gamma, \mathbb{F}_p}$ and $\Pi_{\Gamma, \mathbb{F}_q}$ linear schemes for Γ over $\mathbb{F}_p, \mathbb{F}_q$ respectively. Then $CNF_{\Gamma, \mathbb{F}_p}$ is not convertible to $\Pi_{\Gamma, \mathbb{F}_q}$ for secret domain $S = \{0, 1\}$ (that is, we do not care how other secrets are converted).*

The theorem follows almost immediately from a variant of Lemma 1 for different fields p, q which we provide and prove below.

Claim 5. Let $\Pi = (\mathbb{F}_p, M \in \mathbb{F}_p^{m \times \ell}, \rho)$, $\Pi' = (\mathbb{F}_q, M' \in \mathbb{F}_q^{m' \times \ell'}, \rho')$ for a pair of primes $p \neq q$, and let $T \cup \{h\}$ denote a minterm of Γ . Let $\alpha_{T \cup \{h\}}, \alpha'_{T \cup \{h\}}$ be reconstruction functions for $T \cup \{h\}$ in Π and Π' respectively. Assume $L = \text{Rowspan}(M_T) \cap \text{Rowspan}(M_h) = \{\mathbf{0}\}$. Then for every conversion scheme g from Π to Π' there exists a sequence $\mathbf{r}_1, \dots, \mathbf{r}_i, \dots \in \mathbb{Z}^\ell$ and constant $c \in \mathbb{Z}$ such that (1) $(\alpha'_h)^\top g_h(M_h \mathbf{r}_i \bmod p) \equiv i + c \pmod{q}$; (2) $(\alpha_h)^\top M_h \mathbf{r}_i \equiv i \pmod{p}$ for all $i \in \mathbb{N}^+$, and (3) $\langle \mathbf{r}_i, \mathbf{e}_1 \rangle \bmod p \in \{0, 1\}$. We conclude that such a g does not exist if $p \neq q$.

Proof. Let us consider some $\mathbf{r}_1, \mathbf{r}_2 \in \mathbb{Z}^\ell$ such that

$$M_T \mathbf{r}_1 \equiv M_T \mathbf{r}_2 \pmod{p}; \quad M_h \mathbf{r}_2 \equiv M_h \mathbf{r}_1 + 1 \pmod{p}, \quad (13)$$

and such that $\mathbf{r}_1$ encodes $s = 0$ ($\langle \mathbf{e}_1, \mathbf{r}_1 \rangle \bmod p = 0$) and $\mathbf{r}_2$ encodes $s = 1$. For simplicity, let us fix $\mathbf{r}_1 = \mathbf{0}$. Since $(\alpha_h)^\top M_h$ is linearly independent of M_T , and the fact that $T \cup \{h\}$ is a minterm, such $\mathbf{r}_1, \mathbf{r}_2$ exist. Assume a conversion g between the schemes exists. By Equation (13) and correctness we have that

$$(\alpha_h)^\top M_h \mathbf{r}_2 - (\alpha_h)^\top M_h \mathbf{r}_1 \equiv 1 \pmod{p}.$$

By correctness of Π' and g , and locality of g , we have

$$(\alpha'_h)^\top g_h(M_h \mathbf{r}_1 \bmod p) + (\alpha'_T)^\top g(M_T \mathbf{r}_1 \bmod p) \equiv 0 \pmod{q},$$

¹² For instance, the $(\lceil n/2 \rceil + 1, n)$ -threshold access structure.

$$(\alpha'_h)^\top g_h(M_h \mathbf{r}_2 \bmod p) + (\alpha'_T)^\top g(M_T \mathbf{r}_2 \bmod p) \equiv 1 \pmod{q}.$$

We conclude that

$$(\alpha'_h)^\top g_h(M_h \mathbf{r}_2) - (\alpha'_h)^\top g_h(M_h \mathbf{r}_1) \equiv 1 \pmod{q} \quad (14)$$

We use the following technical observation, to be proven in the sequel.

Observation 27. $(\alpha'_h)^\top g_h(M_h \mathbf{r} \bmod p) \bmod q$ depends only on $(\alpha_h)^\top M_h \mathbf{r} \bmod p$.

Thus, for brevity, we replace $(\alpha'_h)^\top g_h(M_h \mathbf{r} \bmod p) \bmod q$ with $(\alpha'_h)^\top g_h((\alpha_h)^\top M_h \mathbf{r} \bmod p) \bmod q$. Now, consider $\mathbf{r}'_1$ and $\mathbf{r}'_2$ such that $(\alpha_h)^\top M_h \mathbf{r}'_1 \equiv (\alpha_h)^\top M_h \mathbf{r}_2 \bmod p$, $\langle \mathbf{r}'_1, \mathbf{e}_1 \rangle \bmod p = 0$, $(\alpha_h)^\top M_h \mathbf{r}'_2 \equiv (\alpha_h)^\top M_h \mathbf{r}'_1 + 1 \bmod p$, and $(\alpha_T)^\top M_T \mathbf{r}'_1 \equiv (\alpha_T)^\top M_T \mathbf{r}'_2 \bmod p$. Such $\mathbf{r}'_1, \mathbf{r}'_2$ exist by arguments similar to the above, and we have

$$(\alpha_h)^\top M_h \mathbf{r}'_2 - (\alpha_h)^\top M_h \mathbf{r}'_1 = 1 \bmod p.$$

By correctness and locality, we have

$$(\alpha'_h)^\top g_h((\alpha_h)^\top M_h \mathbf{r}'_1 \bmod p) + (\alpha'_T)^\top g_T((\alpha_T)^\top M_T \mathbf{r}'_1 \bmod p) \equiv 0 \pmod{q},$$

$$(\alpha'_h)^\top g_h((\alpha_h)^\top M_h \mathbf{r}'_2 \bmod p) + (\alpha'_T)^\top g_T((\alpha_T)^\top M_T \mathbf{r}'_1 \bmod p) \equiv 1 \pmod{q}.$$

Concluding by correctness for $s = 0, 1$

$$(\alpha'_h)^\top g_h((\alpha_h)^\top M_h \mathbf{r}'_2 \bmod p) - (\alpha'_T)^\top g_T((\alpha_h)^\top M_h \mathbf{r}'_1 \bmod p) \equiv 1 \pmod{q} \quad (15)$$

By $(\alpha_h)^\top M_h \mathbf{r}'_1 \equiv (\alpha_h)^\top M_h \mathbf{r}_2 \bmod p$ and locality, we have

$$(\alpha'_h)^\top g_h((\alpha_h)^\top M_h \mathbf{r}'_2 \bmod p) - (\alpha'_h)^\top g_h((\alpha_h)^\top M_h \mathbf{r}_2 \bmod p) \equiv 1 \pmod{q}. \quad (16)$$

Combining Equation (14) with Equation (16), we conclude that

$$(\alpha'_h)^\top g_h((\alpha_h)^\top M_h \mathbf{r}'_2 \bmod p) - (\alpha'_h)^\top g_h((\alpha_h)^\top M_h \mathbf{r}_1 \bmod p) \equiv 2 \pmod{q}.$$

Generally, proceeding in a similar manner, for $i \geq 0$ (where $\mathbf{r}_b^{(0)} = \mathbf{r}_b$, $\mathbf{r}_b^{(1)} = \mathbf{r}'_b$ and $\mathbf{r}_2^{-1}$ is defined as $\mathbf{r}_1$ for convenience) we conclude that

$$(\alpha'_h)^\top g((\alpha_h)^\top M_h \mathbf{r}_2^{(i)} \bmod p) \equiv i + 1 + (\alpha'_h)^\top g((\alpha_h)^\top M_h \mathbf{r}_1 \bmod p) \pmod{q} \quad (17)$$

This proves the claim, taking $\mathbf{r}_2, \mathbf{r}'_2, \mathbf{r}_2^{(2)}, \dots$ as the required sequence. This holds, as for $c \equiv (\alpha'_h)^\top g((\alpha_h)^\top M_h \mathbf{r}_1 \bmod p) \pmod{q}$, vectors $\mathbf{r}_2^{(-1)}, \mathbf{r}_2, \mathbf{r}'_2, \mathbf{r}_2^{(2)}, \dots$ correspond to values $c, (c+1) \bmod q, (c+2) \bmod q, \dots$ of $(\alpha'_h)^\top g((\alpha_h)^\top M_h \mathbf{r} \bmod p)$, and values $0, 1, 2, 3, \dots$ of $(\alpha_h)^\top M_h \mathbf{r} \bmod p$.

Finally, we prove that g does not exist for $p \neq q$. It holds that $(\alpha'_h)^\top g_h((\alpha_h)^\top M_h \mathbf{r}_{p+1} \bmod p) - (\alpha'_h)^\top g_h((\alpha_h)^\top M_h \mathbf{r}_1 \bmod p) \equiv p \not\equiv 0 \pmod{q}$ since $p \neq q$. On the other hand, $(\alpha_h)^\top M_h \mathbf{r}_{p+1} - (\alpha_h)^\top M_h \mathbf{r}_1 \equiv p \pmod{p}$, which means these values are the same mod p , so by Observation 27 we have $(\alpha'_h)^\top g((\alpha_h)^\top M_h \mathbf{r}_{p+1} \bmod p) \equiv (\alpha'_h)^\top g((\alpha_h)^\top M_h \mathbf{r}_1 \bmod p) \pmod{q}$ – a contradiction. $\square$

Proof of Observation 27. Let $B \cup (\alpha_h)^\top M_h$ be a basis for $\text{Rowspan}(M_h)$ and H be a basis for $\text{Rowspan}(M_T)$. Had $(\alpha'_h)^\top g_h(M_h \mathbf{r} \bmod p) \bmod q$ depend on $B\mathbf{r}$ there would exist a pair $\mathbf{r}_1, \mathbf{r}_2$ of random vectors such that $a = B\mathbf{r}_1 \neq$

$Br_2 = b \pmod{p}$ and $(\alpha_h)^\top M_{T \cup h} \mathbf{r}_1 \equiv (\alpha_h)^\top M_h \mathbf{r}_2 \pmod{p}$, and $(\alpha_T)^\top M_T \mathbf{r}_1 \equiv (\alpha_T)^\top M_T \mathbf{r}_2 \pmod{p}$ are such that overall $\langle \mathbf{e}_1, \mathbf{r}_1 \rangle = \langle \mathbf{e}_1, \mathbf{r}_2 \rangle = 0$, such that $(\alpha'_h)^\top g_h(M_h \mathbf{r}_1 \pmod{p}) \not\equiv (\alpha'_h)^\top g_h(M_h \mathbf{r}_2 \pmod{p}) \pmod{q}$. Thus, by locality on T , had $(\alpha'_h)^\top g_h(M_h \mathbf{r} \pmod{p}) \pmod{q}$ depended on Br , at most one of the secrets recovered from $g(M\mathbf{r}_1)$, $g(M\mathbf{r}_2)$ would equal 0 - a contradiction. $\square$

Now, we are ready to prove Theorem 26.

Proof of theorem 26. Finally, the theorem follows from Claim 5 by observing that for $n > 2$, in $CNF_{\Gamma, \mathbb{F}_p}$ each party p_i gets a subset of independent random vectors over $\mathbb{F}_p$, namely $\mathbf{r}_T$ of each maxterm T such that $i \notin T$. The sets of $\mathbf{r}_H$'s that h holds, vs those T holds. Assume the contrary - that $h \notin H$ and $T \cap H = \emptyset$. In that case $i \notin T \cup H$, contradicting the assumptions that $T \cup H = [n]$ (as T, H are maxterms). This implies that for $CNF_{\Gamma, \mathbb{F}_p}$, $L = \{\mathbf{0}\}$, so the conditions of Claim 5 are indeed satisfied by $\Pi = CNF_{\Gamma, \mathbb{F}_p}$ and $\Pi' = \Pi_{\Gamma, \mathbb{F}_q}$. $\square$

8.3 The specially tailored additive secret-sharing scheme allowing inter-field conversion.

Next, we build the secret-sharing scheme over the field $\mathbb{Z}_p$, and define the conversion function to the field $\mathbb{Z}_q$. The scheme implies some restrictions on the randomness of the dealer, and also is not perfectly secure. However its existence raises a question if there are statistical secure secret-sharing schemes allowing an inter-field conversion, and how small the leakage could be. The other question is about the possibility to convert the original additive scheme over $\mathbb{F}_p$ to a perfectly private scheme over $\mathbb{F}_q$, by using the entire randomness domain of the former. This way, privacy of the converted scheme automatically holds by locality of the conversion scheme. Correctness is the only measure that suffers, with some small probability. This way, we are talking about a conversion scheme between the same pair of perfect schemes Π, Π' , while the conversion scheme itself is statistically correct.

The n -party additive convertible scheme $ADD_{p \rightarrow q}$: **Parameters:** $p \neq q$ are primes such that $q < n/2$. **Sharing algorithm:** (1) the dealer for each $i \in [n]$ samples $r_i \leftarrow \mathbb{Z}_p$. (2) If $\sum_{i=1}^n r_i = kpq + s$, where $s \in \{0, 1\}$ for some k then output $sh_i := r_i$ and terminate. Otherwise go to Step 1. **Conversion function:** Each p_i computes $sh'_i = sh_i \pmod{q}$.

Correctness of this scheme obviously follows from the construction.

Efficiency: The dealer is PPT, since the probability of $r = \sum_{i=1}^n r_i$ be equal to $kpq + s$, for $s \in \{0, 1\}$ and some integer k , where r_i is chosen uniformly at random from $\mathbb{Z}_p$, is polynomial. Hence, on average, the dealer's algorithm does polynomial number of iterations, namely $O(pq/\sqrt{n})$ (see appendix B).

Leakage: The randomness choice of the dealer entails the statistical leakage in this scheme, which is less or equal to $p_{leak} = \frac{1}{p} + o\left(\frac{1}{p^2 n}\right)$ (see appendix B).

Applications: the convertible additive scheme is the basic case for creating the convertible CNF and DNF schemes. However, even directly it could have several applications, similar to applications of *dBits*.

- In generic MPC circuits combining computations in different fields. Say, having a convertible sharing of a random bit s denoted by $[s]_{p \rightarrow q}$, one can convert the sharing of a bit b in $\mathbb{Z}_p$ denoted as $[b]_p$ in a following way: (1) parties compute and recover in MPC mod- p circuit $\Delta = b \oplus s$. (2) Convert $[s]_{p \rightarrow q}$ to $[s]_q$ by a modular reduction and set locally $[b]_p := [s]_q \oplus \Delta$. In a similar way, one can convert $[b]_q$ to $[b]_p$ by comparing $[b]_q$ to $[s]_q$.
- The set of random bits simultaneously shared in different fields allows to perform the same constant-round random permutation in arrays shared in two different fields, by applying series of $\log \ell$ oblivious switches, in $\log \log \ell$ oblivious rounds, where ℓ is the length of the permuted array. This also is a useful primitive for combined circuits.
- It is the useful type of correlated randomness for arithmetic garbled circuits [2,6,19,23], and for MPC-friendly symmetric LPN primitives from alternating moduli [8,11,12].

We leave the existence of practical inter-field convertible secret-sharing schemes with the statistical, or even computational security, as the open question for the future research.

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A Additional details about linear evolving secret-sharing

In this appendix we describe some known constructions as instances of our framework of linear evolving secret-sharing schemes.

First, we describe the evolving linear secret-sharing for the evolving undirected st-connectivity access structure family [16], presented as an IMSP.¹³

Example 1. Consider an evolving undirected st-connectivity access structure specified by a graph $(G(V, E), u_s, u_t)$, as in Definition 13. We construct an IMSP $(\mathbb{F}_2, M, \rho)$ for it as follows (giving rise to a secret-sharing scheme for secrets in $S = \mathbb{F}_2$). In the randomness vector $\mathbf{r} = (r_0 = s, r_1, \dots)$ each element $r_i, i \geq 1$ is associated with $v_i \in V \setminus \{u_s, u_t\}$. For $v \in V$, let $v' = v$ if $v \neq u_t$, and $v' = s$ otherwise. We have a row in M for each party $(u, v) \in E$. For (u, v) where $u \neq u_s$, $M[(u, v)]$ satisfies $M[(u, v), u] = 1$, $M[(u, v), v'] = 1$ and is 0 elsewhere. If $u = u_s$, $M[(u, v), v'] = 1$, and is 0 elsewhere, where $v' = v$ if $v \neq u_t$ and $v' = s$ otherwise.

For details, for the infinite undirected graph in Figure 1 that defines specific evolving undirected st-connectivity access structure, we construct an IMSP over $\mathbb{F}_2$. We obtain the submatrix contributing to the IMSP.

$$\begin{pmatrix} \text{Edges} \\ (u_s, r_1) \\ (r_1, r_2) \\ (r_1, r_3) \\ (r_3, r_4) \\ (r_5, u_t) \end{pmatrix} \begin{pmatrix} s & r_1 & r_2 & r_3 & r_4 & r_5 & \cdots \\ 0 & 1 & 0 & 0 & 0 & 0 & \cdots \\ 0 & 1 & 1 & 0 & 0 & 0 & \cdots \\ 0 & 1 & 0 & 1 & 0 & 0 & \cdots \\ 0 & 0 & 0 & 1 & 1 & 0 & \cdots \\ 1 & 0 & 0 & 0 & 0 & 1 & \cdots \end{pmatrix}$$

¹³ Linear secret-sharing schemes were previously implicitly used in the evolving secret-sharing literature, using the term ‘linear’ without a formal definition, which we first provide here.

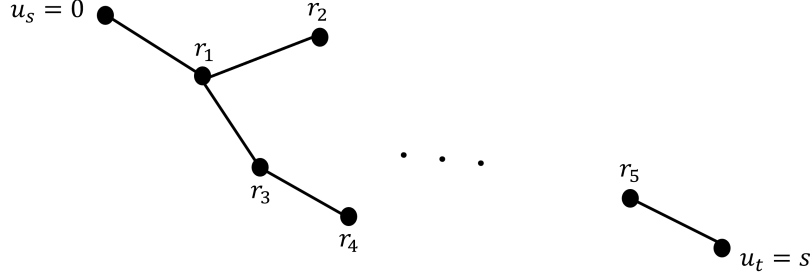


Fig. 1. An example for an infinite undirected graph.

One class of linear schemes falling into our framework in the general GIDT-based constructions.

In the following example we present the constructing for the evolving k -threshold access structure that described in Definition 12 with the scheme $\Pi_{\text{Lin-GIDT}}$.

Example 2. We construct an IMSP for k -threshold over $\mathbb{F}_2$. We use a GIDT for k -threshold, where every node u in the tree ‘remembers’ the number of parties, $\#u$, that have arrived so far (and was still smaller than k). Thus, the predicate on every edge (u, v) , where u is in layer i of the tree, is $(\#v - \#u, |G_i|)$ -threshold. The threshold predicate for (u, v) are then implemented by Shamir over sufficiently large extensions of the field $S = \mathbb{F}_2$, among the parties in G_i . The resulting scheme may be viewed as a linear scheme over $\mathbb{F}_2$ (that is, a scheme induced by a suitable IMSP), by considering multiplication over an extension field as several linear operations over the base field $\mathbb{F}_2$. Let us spell out the first two generations of resulting scheme for $k = 3$. Note that the generation choice is not optimized for share complexity here, and is chosen for presentation purposes. Specifically, we choose $|G_1| = 3, |G_2| = 7$.

We show how to construct the relevant submatrix contributing to the IMSP for the red, blue and green edge in Figure 2.

For the red edge, each of the parties that belongs to generation 1 gets r_1 . This can be seen in the first part of Table 1.

For the blue edge we realize 2-out-of-3 Shamir scheme with the secret r_2 . The smallest possible extension field is $\mathbb{F}_4$. We get the following matrix for that Shamir:

$$\begin{bmatrix} 1 & 1 \\ 1 & 1+x \\ 1 & x \end{bmatrix} \begin{bmatrix} r_2 \\ a_1 + b_1x \end{bmatrix}.$$

In order to represent the elements over $\mathbb{F}_2$, we use modulo the irreducible polynomial $1 + x + x^2$. This can be seen in the second part of Table 1. The blue cells

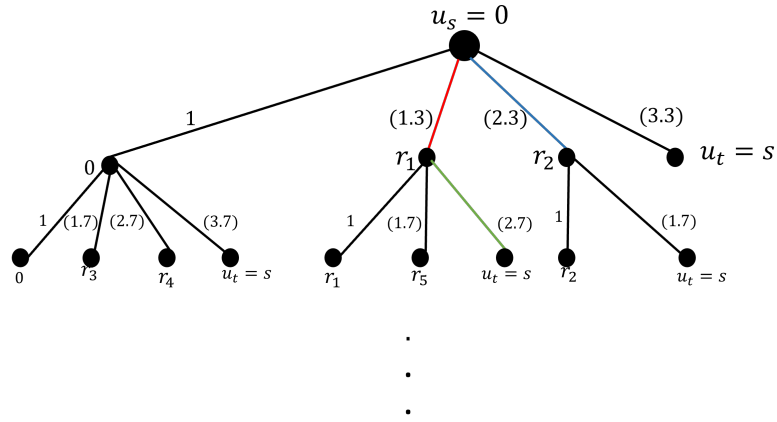


Fig. 2. The GIDT for 3-threshold access structure and generations G_1, G_2 , such that $|G_1| = 3, |G_2| = 7$. The predicates on the edges (k, n) denote as k -out-of- n Shamir scheme.

in Table 1 mark one element in $\mathbb{F}_4$ which is now shown in the 2×2 matrix in $\mathbb{F}_2$.

For the green edge we realize 2-out-of-7 Shamir scheme with the secret $r_2 - s$. The smallest possible extension field is $\mathbb{F}_8$. Like the previous case, we represent the elements over $\mathbb{F}_2$ modulo the irreducible polynomial $1 + x + x^3$. This can be seen in the third part of Table 1. The green cells in Table 1 mark one element in $\mathbb{F}_8$ which is now shown in the 3×3 matrix in $\mathbb{F}_2$.

s	r_1	a_1	b_1	r_2	a_2	b_2	c_2	a_3	b_3	c_3	$\dots$
0	1	0	0	0	0	0	0	0	0	0	$\dots$
0	1	0	0	0	0	0	0	0	0	0	$\dots$
0	1	0	0	0	0	0	0	0	0	0	$\dots$
0	0	1	0	1	0	0	0	0	0	0	$\dots$
0	0	0	1	0	0	0	0	0	0	0	$\dots$
0	0	1	1	1	0	0	0	0	0	0	$\dots$
0	0	1	0	0	0	0	0	0	0	0	$\dots$
0	0	0	1	1	0	0	0	0	0	0	$\dots$
0	0	1	1	0	0	0	0	0	0	0	$\dots$
1	1	0	0	0	1	0	0	1	0	0	$\dots$
0	0	0	0	0	0	1	0	0	1	0	$\dots$
0	0	0	0	0	0	0	0	1	0	1	$\dots$
1	1	0	0	0	0	1	0	0	0	1	$\dots$
0	0	0	0	0	0	0	1	1	1	0	$\dots$
0	0	0	0	0	1	1	0	0	1	1	$\dots$
1	1	0	0	0	0	0	1	0	1	1	$\dots$
0	0	0	0	0	1	1	0	1	1	1	$\dots$
0	0	0	0	0	0	1	1	1	0	1	$\dots$
1	1	0	0	0	0	1	1	0	1	0	$\dots$
0	0	0	0	0	1	1	1	0	0	1	$\dots$
0	0	0	0	0	1	0	1	1	1	0	$\dots$
1	1	0	0	0	1	1	1	1	1	0	$\dots$
0	0	0	0	0	1	0	1	0	1	1	$\dots$
0	0	0	0	0	1	0	0	1	1	1	$\dots$
1	1	0	0	0	1	0	1	1	1	1	$\dots$
0	0	0	0	0	1	0	0	1	0	1	$\dots$
0	0	0	0	0	0	1	0	1	0	0	$\dots$

Table 1. The obtained submatrix contributing to the IMSP for the red, blue and green edge in Figure 2. The $\dots$ in all depicted rows stand for 0's from that point on.

B Characteristics of the convertible inter-field additive scheme $ADD_{p \rightarrow q}$.

In order to calculate the main characteristics of the leaked secret-sharing scheme $ADD_{p \rightarrow q}$ given in section 8.3 we consider the probability distribution of the value $r = \sum_{i=1}^n r_i$, where r_i is chosen uniformly at random from $\mathbb{Z}_p$. As the sum of independent uniformly distributed variables, according to the Central Limit Theorem, it tends to the normally distributed value with the mean $\mu_r =$

$(p-1)n/2$ and variance $\sigma_r^2 = (p^2-1)n/12$. The closeness of the distribution of r to the normal distribution $\mathcal{N}(\mu_r, \sigma_r^2)$ is defined by the Berry-Esseen inequality:

$$\delta_{BE} = |F_n(r) - \mathcal{N}(\mu_r, \sigma_r^2)| \leq \frac{C\rho}{\sigma^3\sqrt{n}}, \quad (18)$$

where $F_n(r)$ and $\mathcal{N}(\mu_r, \sigma_r^2)$ are cumulative distribution functions for r and the normal distribution respectively, σ and ρ are the deviation and the third moment of the distribution of r_i 's, and $C < 0.4748$ is a constant, $\sigma = \sqrt{\frac{(p^2-1)}{12}}$, and $\rho = \frac{(p^2-1)^2}{32p}$. We can accept the estimation $\delta_{BE} \leq \frac{3C\sqrt{p^2-1}}{2p\sqrt{2n}} < \frac{3C}{2\sqrt{2n}} < \frac{0.51}{\sqrt{n}}$. Therefore, for the probability $p_r(x)$ of r to be equal to some x for $0 \leq x \leq (p-1)n$, the estimation bias Δ_{BE} in comparison to the normally distributed variable, is

$$\Delta_{BE} = \frac{\delta_{BE}}{(p-1)n+1} < \frac{0.51}{pn\sqrt{n}}. \quad (19)$$

The estimation $p_{\mathcal{N}}(x)$ for the probability $\Pr[r = x]$ obtained from the normal distribution is

$$p_{\mathcal{N}}(x) = \frac{1}{2} \left(\operatorname{erf} \left(\frac{x+1-\mu_r}{\sqrt{2}\sigma_r} \right) - \operatorname{erf} \left(\frac{x-\mu_r}{\sqrt{2}\sigma_r} \right) \right), \quad (20)$$

where $\operatorname{erf}(x) = \frac{2x}{\sqrt{\pi}} - \frac{2x^3}{3\sqrt{\pi}} + o(x^5)$ for $x \rightarrow 0$.

Probability of a successful termination. The sharing of $s \in \{0, 1\}$ is obtained successfully if r gets in one of the windows $\{w_k, w_k + 1\}$ where $w_k = kpq$, and $k \in \{0, \dots, k_{\max}\}$ for $k_{\max} = \lfloor \frac{(p-1)n}{pq} \rfloor$. Next we compute the probability of a successful termination of the dealer's algorithm in one iteration. For this, let $w_k = \mu_r + i$, and then from (20) for $i \geq 0$ it follows that

$$p_{\mathcal{N}}(\mu_r + i) = \frac{1}{2} \left(\operatorname{erf} \left(\frac{i+1}{\sqrt{2}\sigma_r} \right) - \operatorname{erf} \left(\frac{i}{\sqrt{2}\sigma_r} \right) \right) = \quad (21)$$

$$= \frac{i+1}{\sqrt{2\pi}\sigma_r} - \frac{i}{\sqrt{2\pi}\sigma_r} - \frac{(i+1)^3}{6\sqrt{2\pi}\sigma_r^3} + \frac{i^3}{6\sqrt{2\pi}\sigma_r^3} + o(\sigma_r^{-5}) = \quad (22)$$

$$= \frac{1}{\sqrt{2\pi}\sigma_r} \left(1 - \frac{i(i+1)}{2\sigma_r^2} \right) + o(\sigma_r^{-5}), \quad (23)$$

and, similarly,

$$p_{\mathcal{N}}(\mu_r + i + 1) = \frac{1}{\sqrt{2\pi}\sigma_r} \left(1 - \frac{(i+1)(i+2)}{2\sigma_r^2} \right) + o(\sigma_r^{-5}). \quad (24)$$

From (21) and (24) it follows that

$$\begin{aligned} \Pr[r \in \{\mu_r + i, \mu_r + i + 1\}] &= p_{\mathcal{N}}(\mu_r + i) + p_{\mathcal{N}}(\mu_r + i + 1) \pm 2\Delta_{BE} = \\ &= \frac{1}{\sqrt{2\pi}\sigma_r} \left(2 - \frac{i+1}{\sigma_r^2} \right) \pm 2\Delta_{BE} + o(\sigma_r^{-5}). \end{aligned} \quad (25)$$

For negative i the distribution is symmetric, hence in (25) there should be $|i|$ in the bracket.

From the fact that there are $k_{\max} + 1 \approx \frac{n}{q}$ windows, on the distance pq from each other, it follows that

$$\begin{aligned}
\Pr[r \in \{w_k, w_k + 1\} | 0 \leq k \leq k_{\max}] &= 2 \sum_{k=0}^{k_{\max}/2} \Pr[r \in \{\mu_r + kpq, \mu_r + kpq + 1\}] = \\
&= \frac{2\sqrt{2}k_{\max}}{\sqrt{\pi}\sigma_r} - \frac{\sqrt{2}}{\sqrt{\pi}\sigma_r^3} \sum_{k=0}^{k_{\max}/2} (kpq + 1) \pm 2k_{\max}\Delta_{BE} + o(n\sigma_r^{-5}/q) = \\
&= \frac{2\sqrt{2}k_{\max}}{\sqrt{\pi}\sigma_r} - \frac{pqk_{\max}}{2\sqrt{2}\pi\sigma_r^3} \pm 2k_{\max}\Delta_{BE} + o(n\sigma_r^{-5}/q) = \\
&= O\left(\frac{\sqrt{n}}{pq}\right) - O\left(\frac{1}{p^2\sqrt{n}}\right) \pm O\left(\frac{1}{pq\sqrt{n}}\right) + o\left(\frac{1}{p^5qn\sqrt{n}}\right) = O\left(\frac{\sqrt{n}}{pq}\right). \quad (26)
\end{aligned}$$

Leakage in $ADD_{p \rightarrow q}$. In $ADD_{p \rightarrow q}$, corrupt parties, given their shares r_i 's, and the fact that the protocol was terminated successfully, i.e. $r \in \{w_k, w_k + 1\}$ for some $k \in \{0, \dots, \lfloor \frac{(p-1)n}{pq} \rfloor\}$, could learn the information about the shares of honest parties. The maximal leakage is obtained when the number of corrupt parties is $(n-1)$, i.e. there is only one honest party P_i , in case when $r' = r - r_i$ equals to $w_k - (p-1)$ (and then they know that $s = 0$, and $r_i = p-1$), or when $r' = w_k + 1$ (then $s = 1$, and $r_i = 0$). Next we estimate the probability of leakage, i.e. the probability that the shared bit is known to the adversary. This probability is

$$p_{leak} = \Pr[r' \in \{w_k - (p-1), w_k + 1\} | r \in \{w_k, w_k + 1\}] = \quad (27)$$

$$= \frac{\Pr[r \in \{w_k, w_k + 1\} | r' \in \{w_k - (p-1), w_k + 1\}] \Pr[r' \in \{w_k - (p-1), w_k + 1\}]}{\Pr[r \in \{w_k, w_k + 1\}]} \quad (28)$$

Taking into account that r_i is uniformly distributed in $\mathbb{Z}_p$, and from equation (25) it follows that

$$p_{leak} = \frac{1}{p} \cdot \frac{\Pr[r' \in \{w_k - (p-1), w_k + 1\}]}{\frac{\sqrt{2}}{\sqrt{\pi}\sigma_r} + o(\sigma_r^{-3})}, \quad (29)$$

where, from (20) it follows that

$$\begin{aligned}
\Pr[r' \in \{w_k - (p-1), w_k + 1\}] &\leq \frac{1}{2} \left(\operatorname{erf}\left(\frac{w_k - p + 1 - \mu_{r'}}{\sqrt{2}\sigma_{r'}}\right) - \right. \\
&- \operatorname{erf}\left(\frac{w_k - p + 1 - \mu_{r'}}{\sqrt{2}\sigma_{r'}}\right) + \operatorname{erf}\left(\frac{w_k + 2 - \mu_{r'}}{\sqrt{2}\sigma_{r'}}\right) - \operatorname{erf}\left(\frac{w_k + 1 - \mu_{r'}}{\sqrt{2}\sigma_{r'}}\right) \Bigg) + \\
&\quad + 2\Delta_{BE} = \frac{\sqrt{2}}{\sqrt{\pi}\sigma_{r'}} + o(\sigma_{r'}^{-3}),
\end{aligned}$$

Thus,

$$p_{leak} = \frac{1}{p} + o(\sigma_r^{-2}) = \frac{1}{p} + o\left(\frac{1}{p^2 n}\right), \quad (30)$$

and the adversary knows the shared bit s with the probability less or equal to p_{leak} .